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STAGE 3: CUSTOM GEARBOX DESIGN

MECH 360 DESIGN PROJECT

December 8, 2024

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MECH 360 DESIGN PROJECT

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1. Abstract

This report examines several existing gearbox designs and puts forward a possible gearbox design for use in a steam turbine solar thermal power generation plant. To accommodate the intended operating conditions of the turbines and generator, a compound gearbox using 8 steel spur gears to form a 4-stage reduction has been designed to transfer the power to the generator at a nominal speed reduction factor of 13.33. The gearbox is roughly 170 inches in length and is housed within a cast iron casing. The analysis and design of the shafts, gears, and bearings are detailed in this report.

2. Introduction

The purpose of this project is to design a gearbox speed reduction system for a new 30 MW concentrated solar power (CSP) plant proposed by ACS. This smaller CSP plant will use a steam turbine connected to a gearbox, which will, in turn, drive a synchronous generator to inject electricity into the grid. The engineering team will be designing the critical components of the gearbox, including shafts, gears, bearings, and housing, ensuring the system operates efficiently and reliably under varying load conditions.

This work supports the development of renewable energy systems that can provide electricity during night-time hours when photovoltaic systems are not operational. By storing solar energy in molten salt during the day and generating electricity at night, the CSP plant is a more sustainable solution for energy production. The project also emphasizes detailed mechanical design, focusing on the durability and performance of components that will contribute to the long-term reliability of renewable energy infrastructure.

3. Background

Concentrated Solar Power (CSP) is a renewable energy technology that uses mirrors to focus sunlight onto a central receiver, where the heat generated is absorbed by a working fluid, often molten salt. This molten salt is stored in large tanks and can retain heat for extended periods, allowing for the generation of electricity even after the sun has set. CSP systems typically use this heat to produce steam, which drives a turbine connected to a generator, converting thermal energy into electrical energy. Using CSP systems is beneficial because they provide a reliable source of electricity during hours of sunlight and darkness, unlike photovoltaic (PV) systems, which only generate power during daylight.

The requirements of this speed reduction system are similar to other large industrial gear reduction systems, such as speed reducers found in hydropower plants and fossil fuel generating stations.

4. Review of Literature

4.1 Case Study, Sumitomo Paramax 9000 Industrial Gearbox

This design uses a combination of helical and bevel gears to achieve a reduction factor of 14.0. They use ground and case-carburized gears with a 25-degree pressure angle to withstand loading conditions of up to 500 KNm at 5230 horsepower. Extra wide bearings (either tapered, double-row spherical, or cylindrical) are used to support the forces on the shafts. They also spec all keys and keyways for an interference fit to allow the system to handle shock loads and start-and-stop operation better. To prevent loss of lubricant, two seals are used on the input and output shafts to ensure longevity.

This is a useful comparison because it is very close in capability to what the power plant gearbox will require, and it has actionable changes we can incorporate into the design to improve performance.

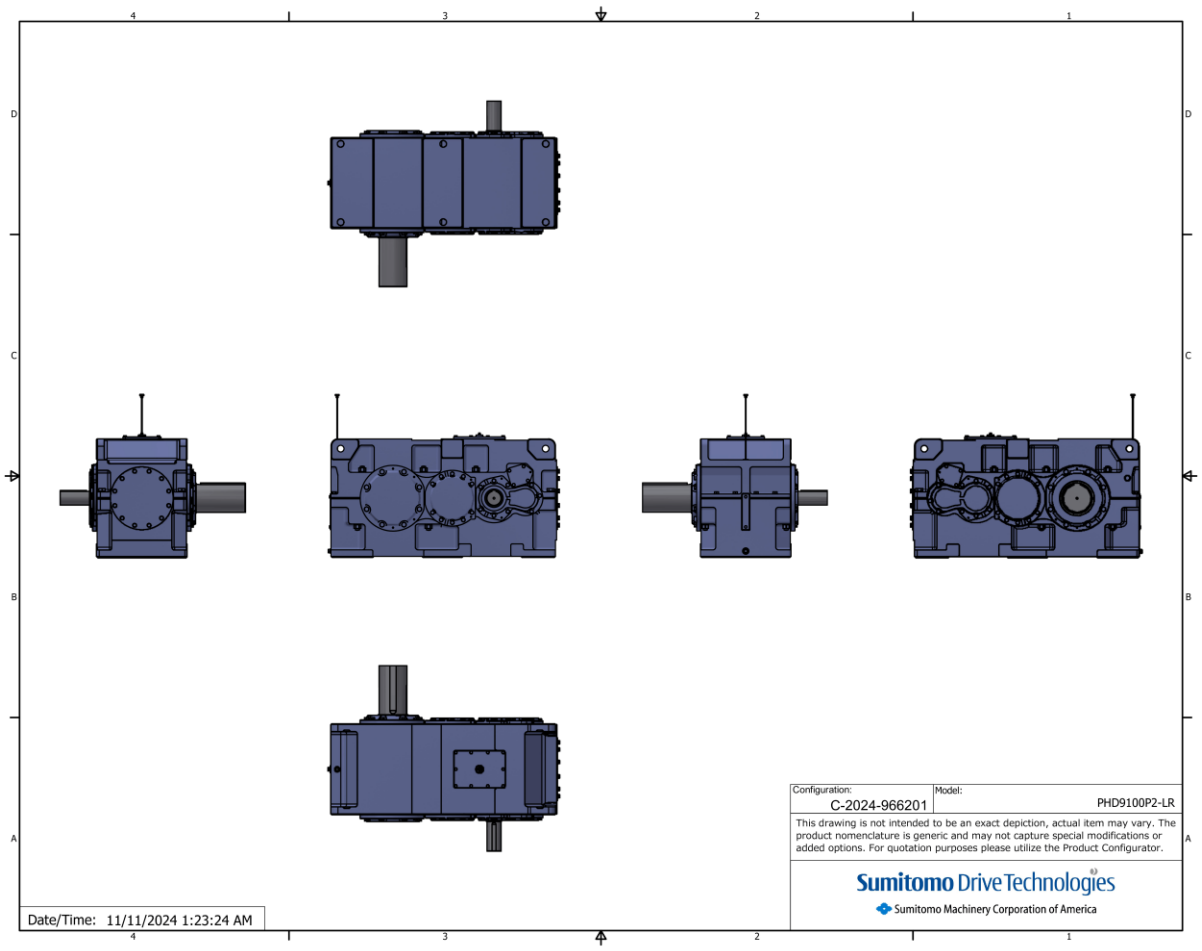


Figure 1: Sumitomo Hansen-P4 Gearbox

4.2 Case Study, Worldwide Electric Corporation Model WINL147-10/1-405TC

This gearbox is a smaller design with a parallel input and output shaft meant for a maximum power throughput of 200 horsepower. It uses a reverted 2-stage design to achieve a reduction factor of 10.24. The unit uses ball bearings and precision ground gears and claims that this reduces vibration and noise during operation. This Design uses a cast iron housing that weighs 915 pounds and retails for $ 15,930 USD.

While simple, this gearbox is functional and shows that a reverted compound geartrain can be commercially successful.

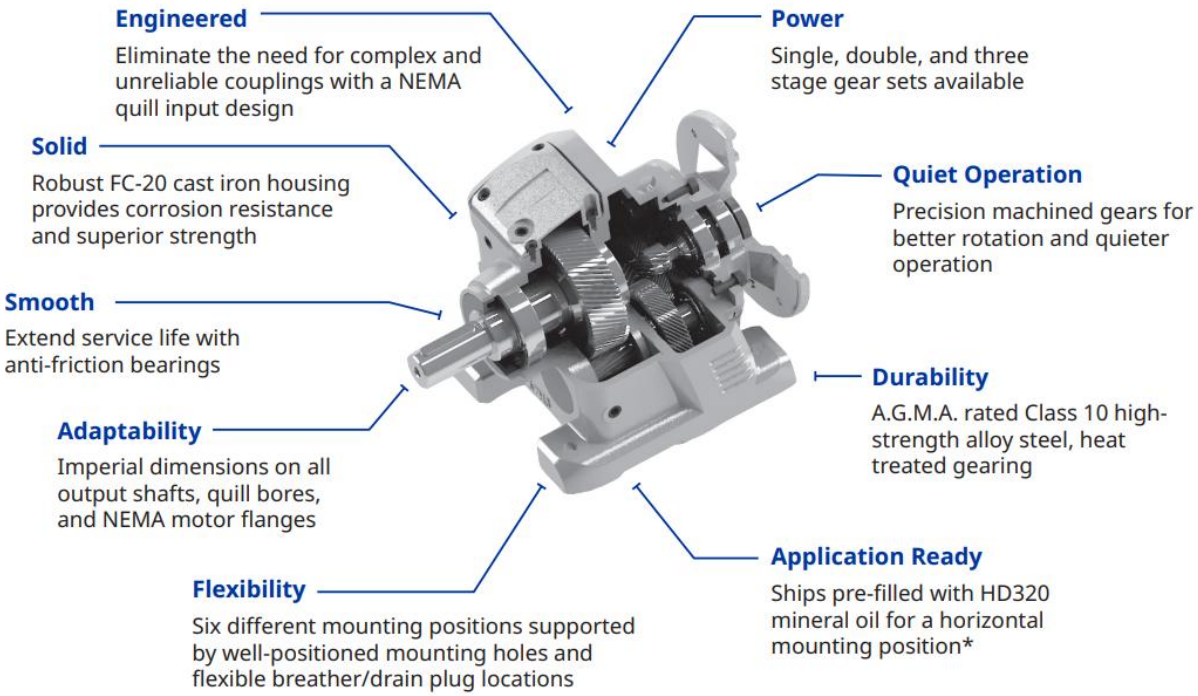


Figure 2: WEC Gearbox

4.3 Market Survey

The cost of renewable energy projects of this scale is relatively high, largely due to recent technology. However, these costs are expected to decrease over time as newer technologies are developed and adopted. Although high-efficiency gearboxes like the 2-stage reduction gearbox may have higher upfront costs (96-97% for the specified design), they offer long-term savings by reducing maintenance expenses and optimizing operational output. This makes them a strategic investment for renewable energy projects (especially CSP plants), providing better performance, reliability and long-term savings throughout the lifespan of the system.

The market for very large gearboxes is made up of three main categories [4] in order of similarity to the current project.

- Power generation and transmission.
- Mills, processing plants, and manufacturing.
- And Earth-moving, mining, and excavation.

In the context of this project, power generation and transmission projects will be considered due to their similarity. In 2016, BC Hydro awarded the Cite C power generation Turbine and Generator contract to Voith Group KGaA for a sum of $464M CAD [6].

The cost to replace a gearbox in mining or construction may be between 50 K$ and 200 K$ USD. [7]

With this in mind, we estimate that a market-acceptable price is somewhere in the 400-to-600-thousand-dollar USD range.

5. List of Design Objectives & Requirements

To ensure the success of the gearbox design, the following table of objectives/ requirements has been made. Each objective/requirement has a performance measure/target value, and a justification. This will allow for a thorough evaluation of the design.

Objective/ Requirement	Quantifiable Value	Justification
Nominal Rated Electrical Output Power	15 MW per turbine	System is designed to generate 15 MW per turbine, creating a total of 30 MW, to meet the requirement for nighttime energy generation.
Gearbox Efficiency	96-97%	Gearbox efficiency is expected to be 96-97%. Gear stage efficiency calculations should reflect this value.
Turbine Speed	6000 RPM	Gearbox must be able to accommodate this turbine speed.
Synchronous Generator Speed	450 RPM	The gear must accurately achieve this speed, considering efficiencies and gear ratio calculations.
System Design Lifetime	25 years	The system is expected to operate for 25 years, with only minor maintenance. Must take this into account when choosing materials.
Gearbox Layout Design	Designed for two parallel turbine-	The gearbox layout will include two parallel systems, which can handle 15 MW of peak power each.

	gearbox-generator systems	
Gearbox Housing Design	Housing with internal locating features and external mounting points	The gearbox housing must include locating features for internal parts, and mounting points to the steel support frame, allow for a lubrication circulation system and must consider possible maintenance. Also, should be able to assemble with ease.
Shaft Sizes and Alignment	Input/output shafts enter and exit at equal heights above the base	Shaft must be appropriately selected to handle the power and torque requirements. The entry and exit points should be aligned.
Gear Teeth Specification	Detailed specifications required for all gears	Accurate gear teeth specifications are necessary to ensure proper meshing and load distribution. Involute gear profiles will be used. Minimizing wear and ensuring a 25-year lifetime.
Couplers Choice	Appropriate couplers to mate input and output shafts	Couplers must be appropriate to mate with the gearbox's input/output shafts. The choice of couplers will affect the shaft sizing.
Support Bearings	High-load capacity bearings for input, intermediate and output shafts.	Support bearings must be capable of handling the forces generated by the input, intermediate, and output shafts. Reducing the risk of mechanical failure.
Fatigue Analysis	Complete calculations with appropriate safety factors	Fatigue effects due to daily load variations must be considered to prevent failure within the 25-year lifetime. Also, optimize the gearbox's service life.
Lubrication Circulation System	Input/output ports	Must provide input and output ports for a lubrication circulation system.
Assembly and Maintenance Accessibility	Accessible design for maintenance	The gearbox housing and layout should allow for easy assembly and maintenance to minimize downtime and facilitate repairs or part replacements.

Cost Efficiency	Take cost into consideration when choosing materials	Minimize cost while ensuring other requirements are met. Important when considering materials.
Space Optimization	Consider a design that minimizes the amount of space being used	This will lead to space savings.

Table 1: Gearbox Objectives and Requirements

6. Overall Concept Design

The gearbox design will consist of 8 gears arranged in four series connected reverted reduction stages as shown in Figure 3. The rotating output shaft from the turbine will be connected to the input shaft of the speed reducer. This will feed through the reduction stages and will output co-linearly with the input to conserve the form factor of the power plant. The Housing is a two-part cast-iron design with critical features machined post-casting.

The shafts' design aims to optimize the stresses and safety factors on the shafts, as they will be undergoing large forces due to the input torque and gear meshing. This is balanced with a need for methods to constrain elements such as gears and bearings, as well as to transmit torques to and from different gears. The methods we chose to do include keyways to transmit gear torque, as they are simple to implement and can be made large to accommodate the stresses we are experiencing. The design team picked clamping collars and shoulders to handle the axial locating of components as they are significantly stronger than snap rings but can still be installed and removed with relative ease and do not require any extreme features on the shafts, and shoulders are a strong and simple way to ensure the gears/bearings cannot slide off a shaft. The areas of the shaft that are highlighted in Figure 8 represent the location where fatigue analysis will be completed (this is discussed in depth in Section 9.5 of this report).

The gears in the gearbox will transmit large amounts of torque to decrease the given speed of the motor to the desired velocity of the generator. The points of gears that are in mesh will be subject to bending stress and surface stress. These locations were identified as potential critical locations for failure where failure analysis was conducted and are shown in Figure 6.

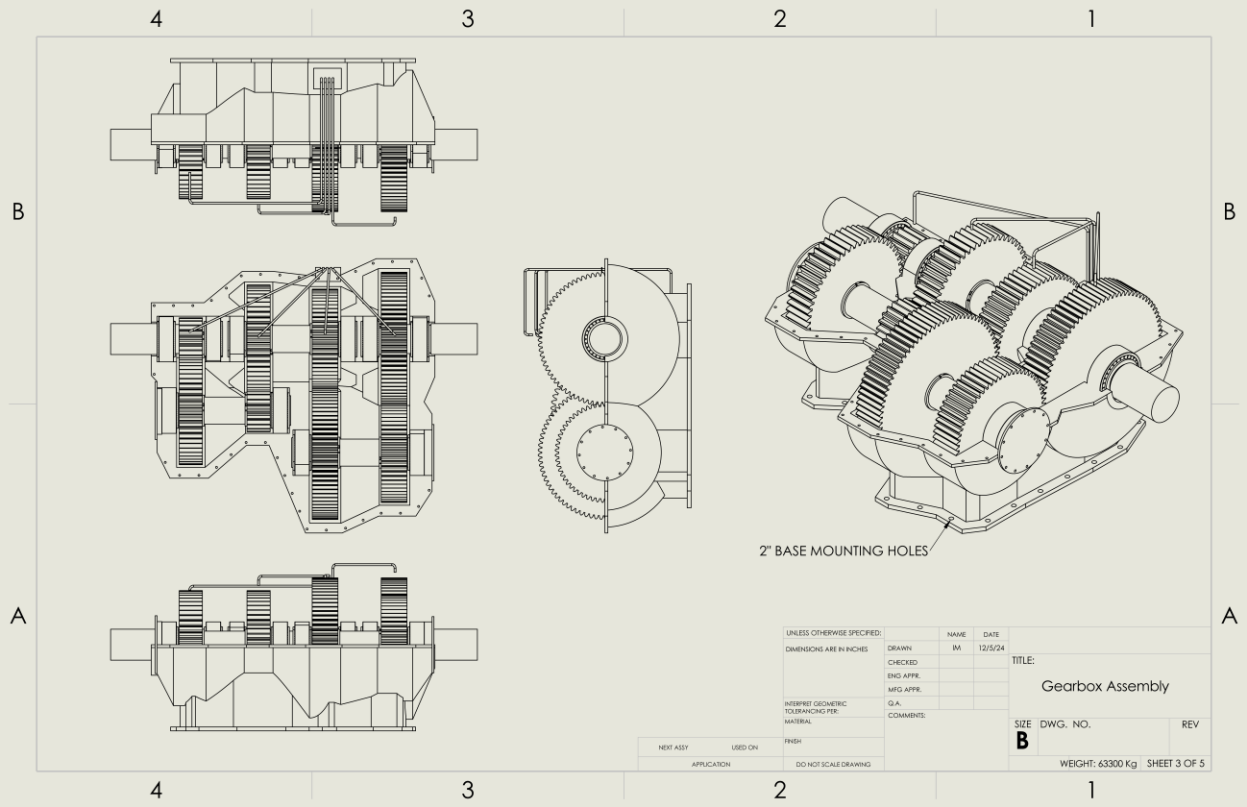


Figure 3: Overview Drawing

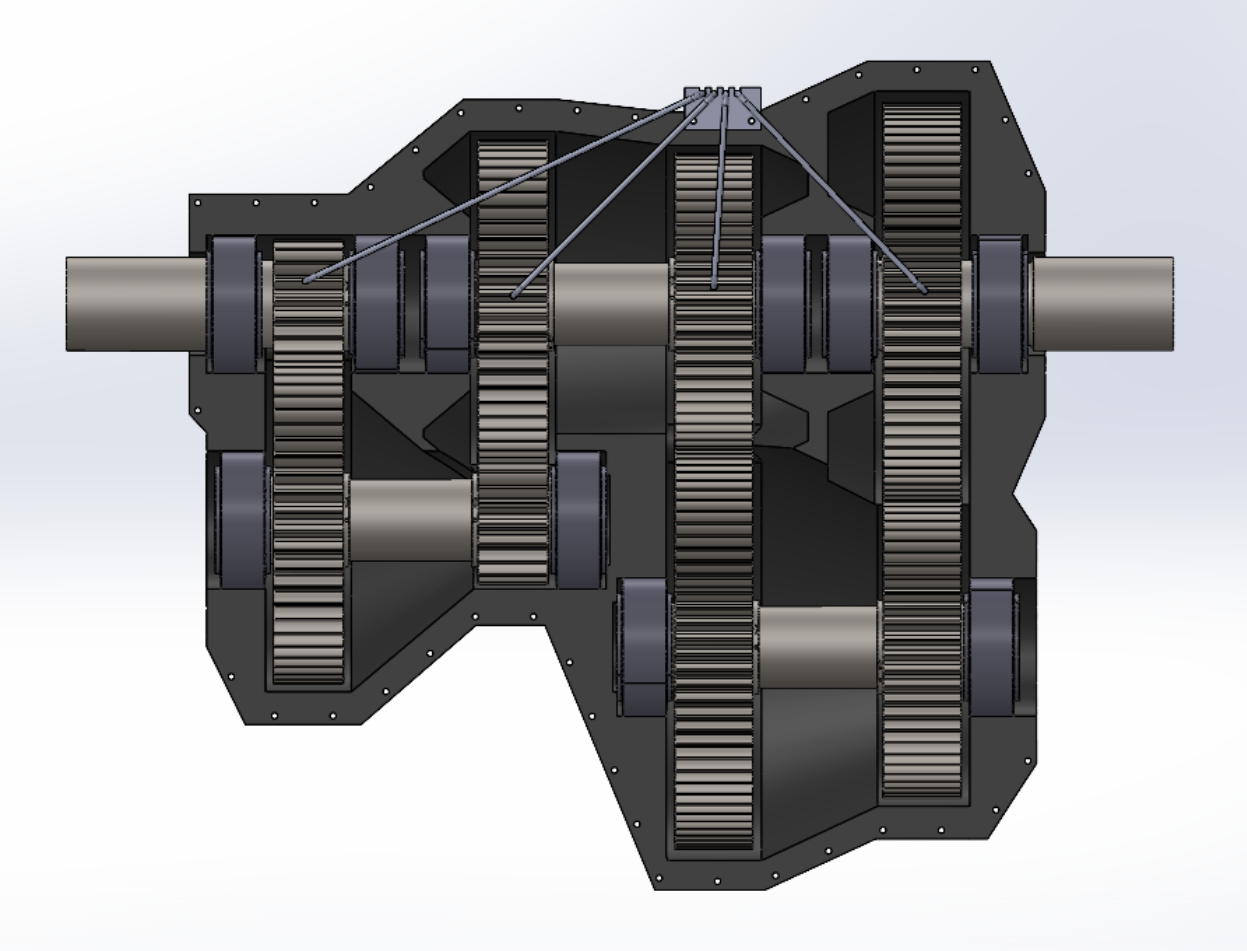


Figure 4: Top View Cross Section

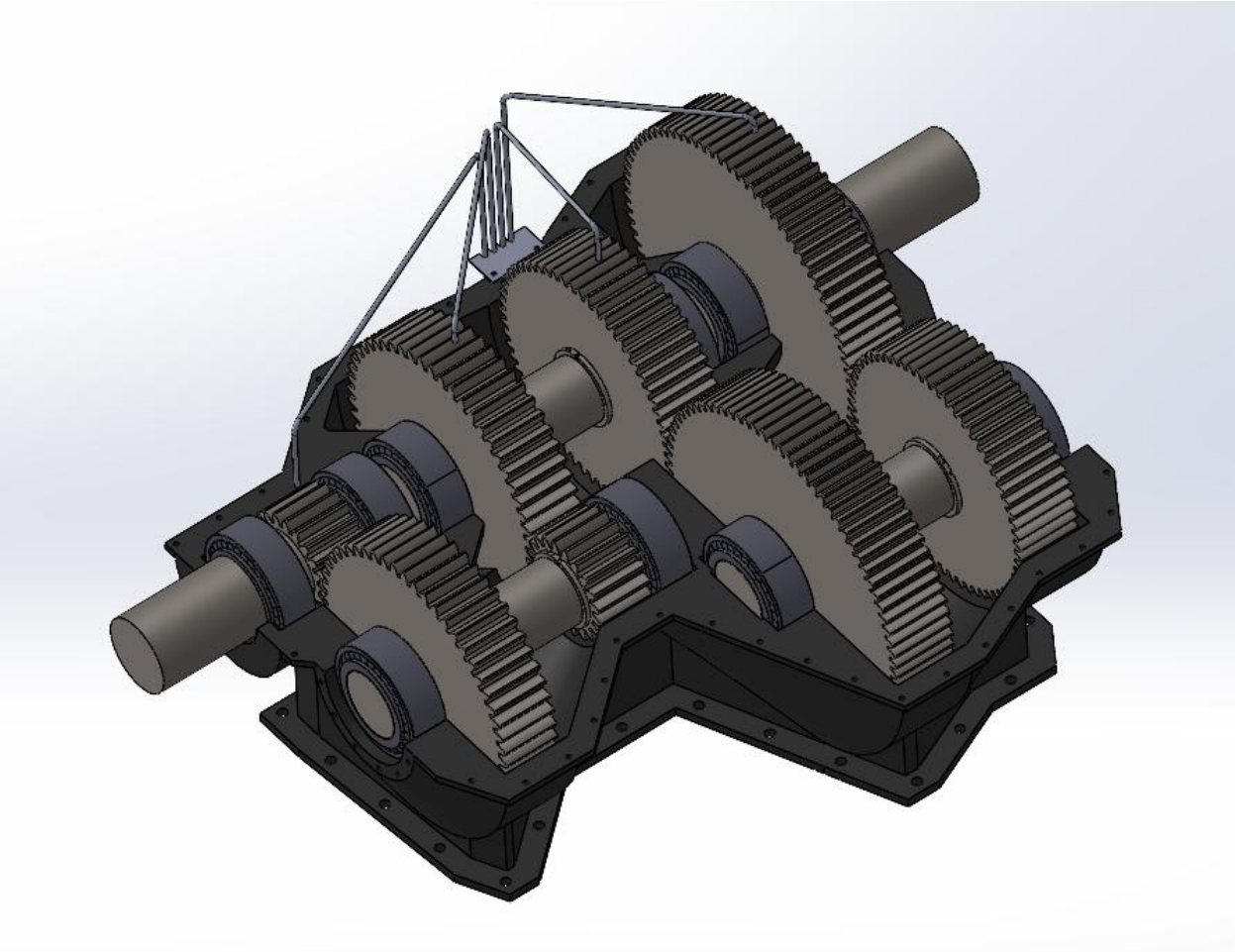


Figure 5: Isometric View

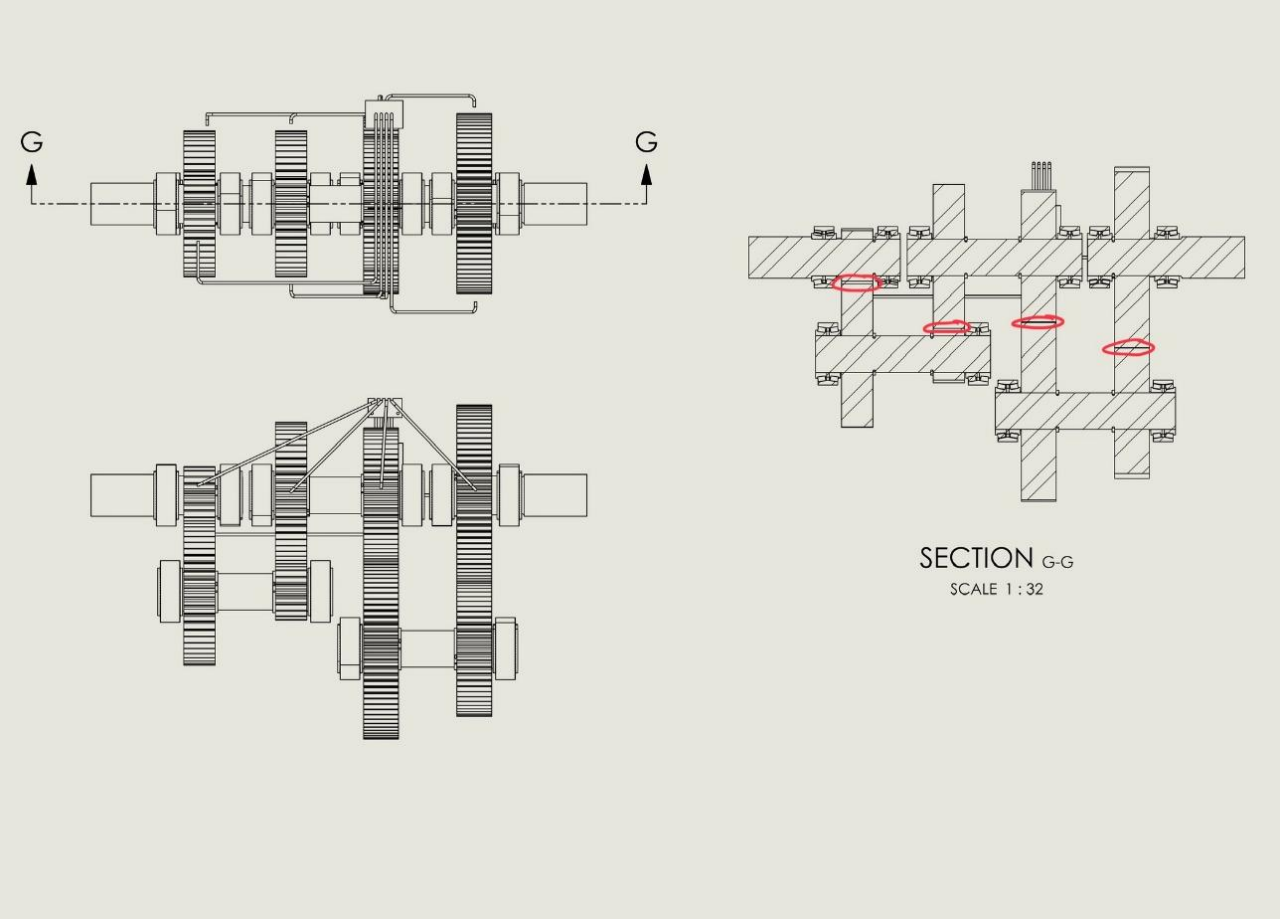


Figure 6: Section View Showing Potential Critical Locations for Failure

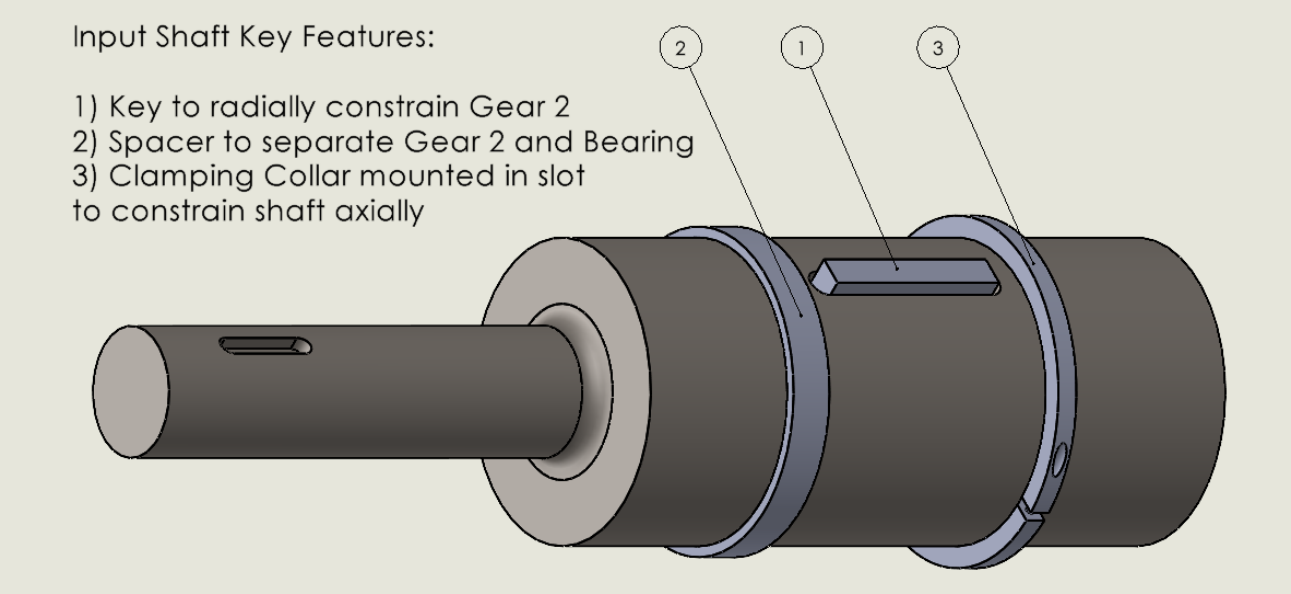


Figure 7: Input Shaft Key Features

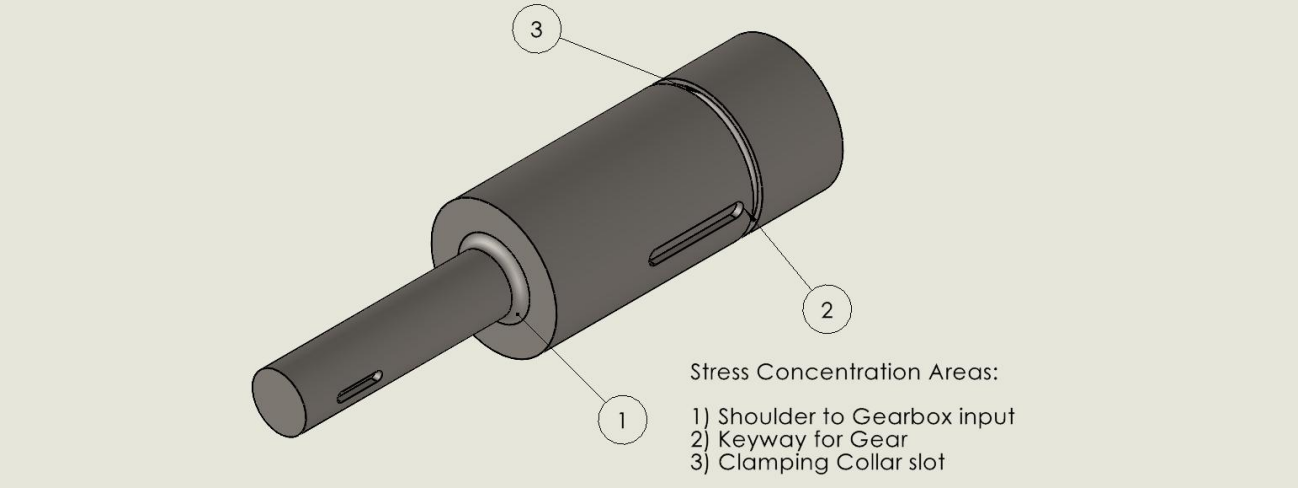


Figure 8: Critical Areas for Fatigue Analysis of Input Shaft

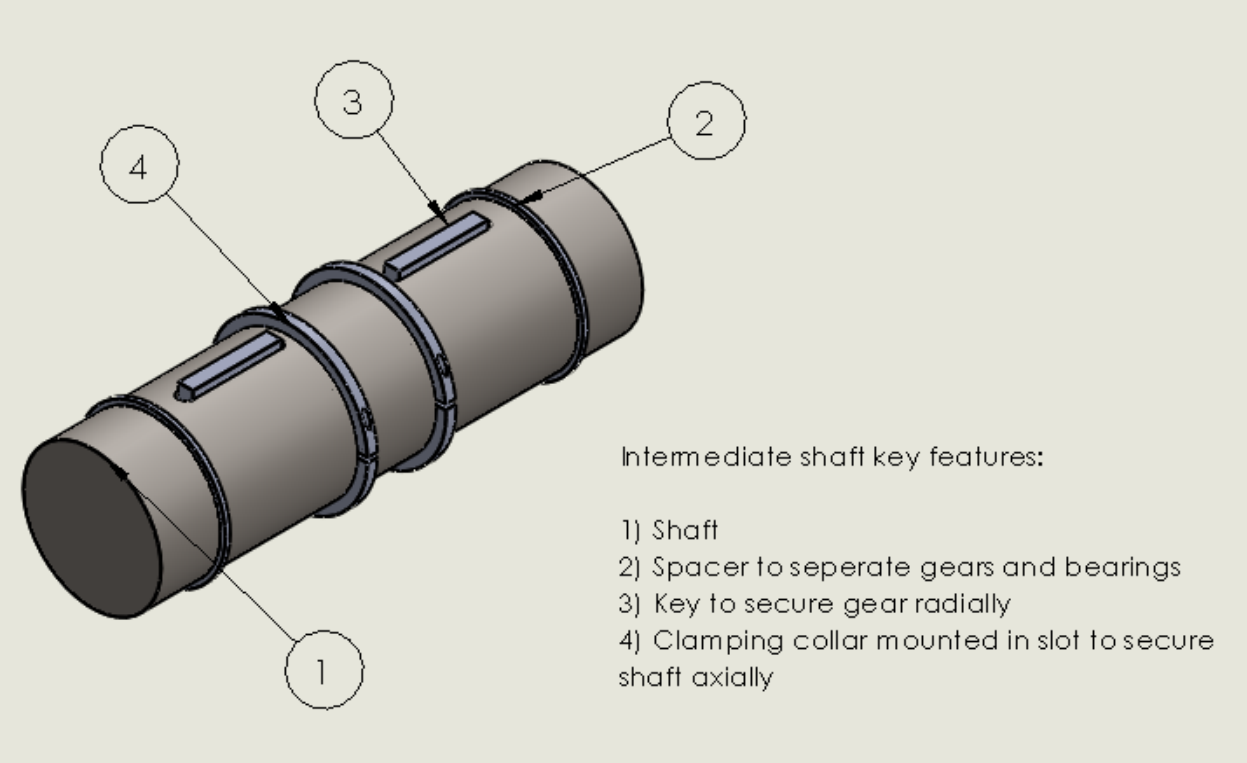


Figure 9: Intermediate Shaft Key Features

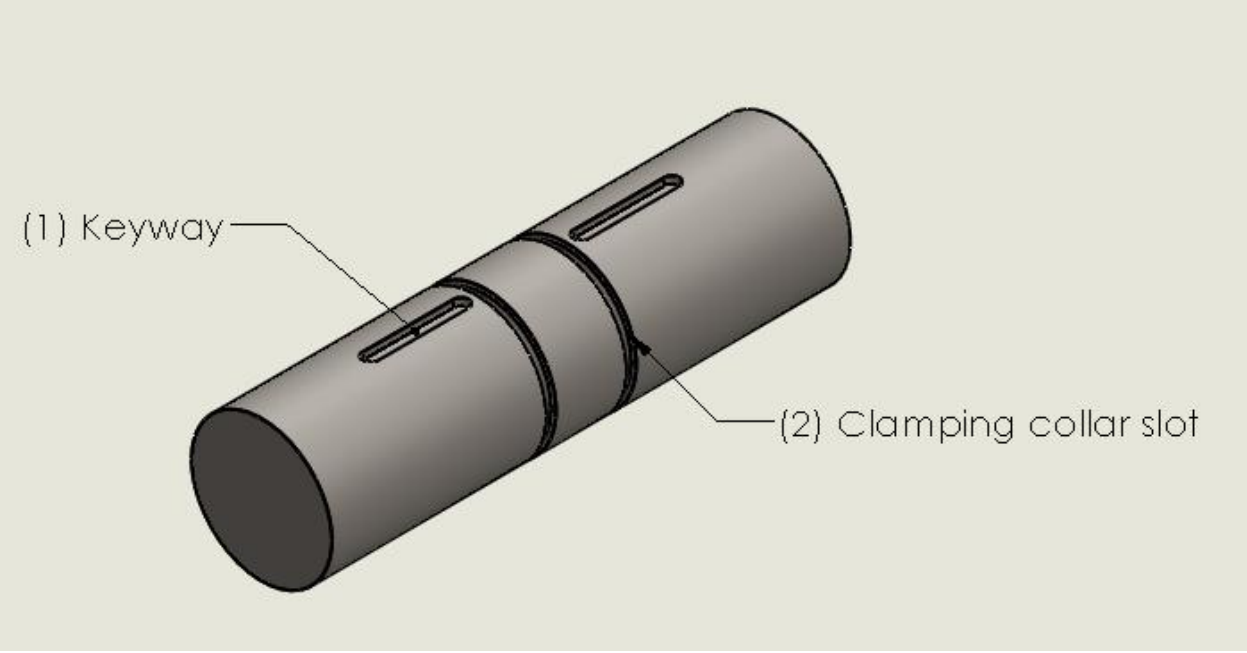


Figure 10: Critical Areas for Fatigue Analysis of Intermediate Shafts

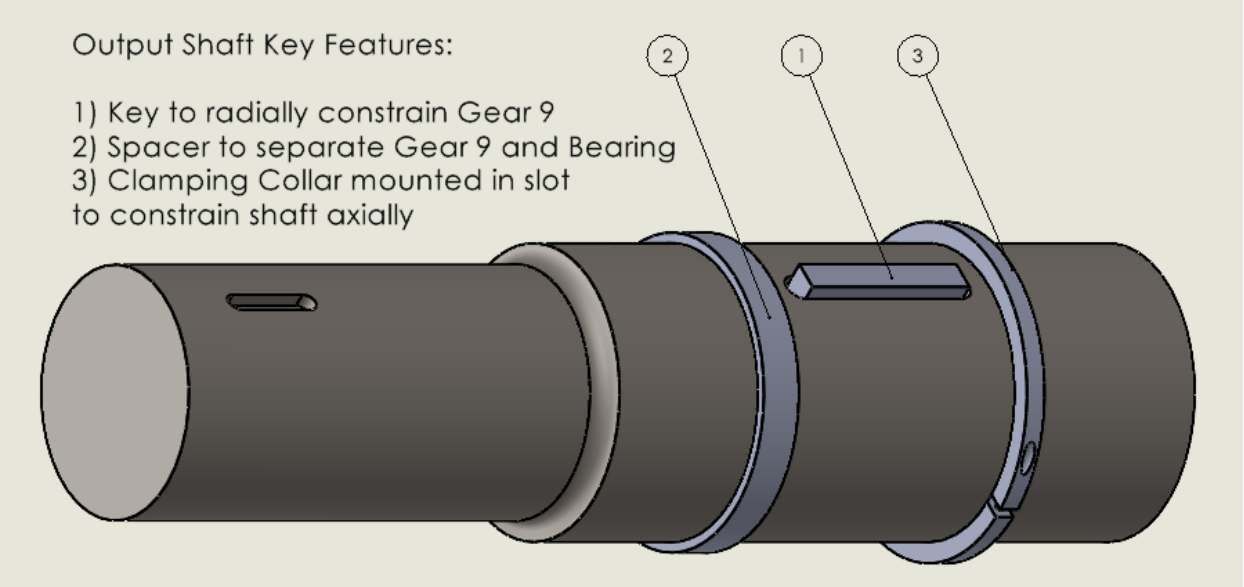


Figure 11: Output Shaft Key Features

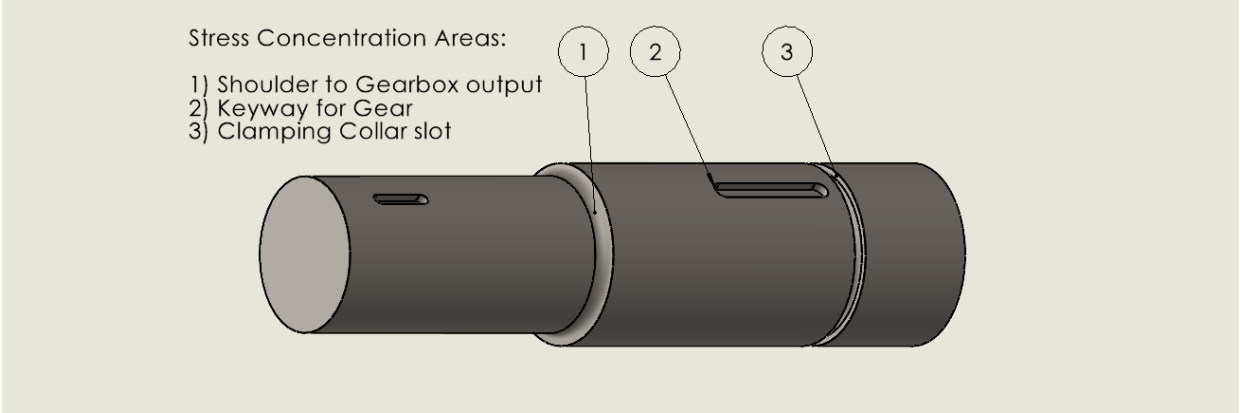


Figure 12: Critical Areas for Fatigue Analysis of Output Shaft

7. Gearbox Element Concept Design

The gearbox was designed to accommodate high torque, fast speeds, and varying loads, while also maintaining efficiency and structural integrity. Coupling used for the input and output shaft are intended to transfer torque, with additional benefits such as misalignment compensation, and

vibration dampening. Calculations were completed to evaluate whether additional couplings were needed to address shaft misalignment. The calculations confirmed the misalignment was minimal therefor no additional couplings were required. Bearings were used throughout the gearbox to support the loads, dampen vibration, allow for smooth motion, and reduce friction.

The assembly sequence will involve sliding the bearings and gears onto their respective shafts before, with the bearings being located by a shoulder on the shaft and the outer housing. The spur gears will be affixed to the shafts with keyways, resulting in no axial stresses. After the shaft components are installed, the shafts can be lowered into the lower half of the casing, being supported by their bearings. Couplings will be used to secure the input shaft to the turbine and the output shaft to the generator. The two housing halves will have slots for a gasket to seal against the lubricants inside the gearbox. The shafts will have a round shoulder on the outside of the housing to accommodate a shaft seal. After the box is sealed and all fasteners are torqued, it can be filled with lubricants and put into service.

8. Speeds and Loads Concept Design

8.1 Gearbox Reduction Ratios

The gearbox is designed as a speed-reducing gearbox, connected between a turbine and a generator. The overall gearbox speed reduction is calculated using the specified turbine and generator speeds.

Turbine speed: 6000 rpm

Generator speed: 450 rpm

$$Overall\ gearbox\ speed\ reduction = \frac{6000\ rpm}{450\ rpm} = 13.\overline{3}$$

The design team decided the overall speed reduction would best be achieved by a 4-stage compound gearbox. For simplicity, the gearbox was designed to have 4 stages, with 2 sets of equal stages.

With gear 2 being the input and gear 9 being the output, the overall speed ratio, $e_{9/2}$ is 13.3.

The team experimented with different combinations of reduction ratios for each stage, with the following combination being the best for reducing bending and surface stresses. The reductions are as follows:

$$e_{\frac{3}{2}} = -2.7$$

$$e_{\frac{4}{3}} = 1$$

$$e_{\frac{5}{4}} = -2.7$$

$$e_{\frac{6}{5}} = 1$$

$$e_{\frac{7}{6}} = -1.35$$

$$e_{\frac{8}{7}} = 1$$

$$e_{\frac{9}{8}} = -1.35$$

Thus:

$$\boldsymbol{e_{\frac{9}{2}}} = e_{\frac{3}{2}} * e_{\frac{5}{4}} * e_{\frac{7}{6}} * e_{\frac{9}{8}} = \mathbf{13.3}$$

This table summarizes gear speed and reductions:

Stage	Speed Ratio	Gear speed [rpm]
1	-2.7	2222
2	-2.7	823
3	-1.35	609
4	-1.35	450

Table 2: Speed Ratios and Gear Speeds

The table above shows each gear stage with its corresponding speed ratio and gear speed. This setup achieves the target speed reduction from input to output.

8.2 Gear Dimensions

The following gear reductions were chosen to achieve the needed speed ratios. A low Pitch Diameter (Pd) was chosen to increase the face width and lower bending stress. Below are sample calculations for stages 1 and 2, along with a summary table of gear dimensions.

Pitch Diameter Sample Calculation:

The pitch diameter (Pd) is calculated for each gear stage, representing the diameter of the pitch circle where force is transferred between meshing gears.

1. Stage 1 and 2: $$P_d = \frac{N_p}{d_p} = \frac{20}{20} = 1.0 \text{ in}^{-1}$$

Base Diameter Sample Calculation:

The base pitch (Pb) is the diameter of the circle at the base of the gear teeth, where the gear's involute profile starts. It is calculated as follows:

For stages 1 and 2: $$P_b = \frac{\pi cos\emptyset}{P_d} = \frac{\pi \cos(25°)}{1.0\ in^{-1}} = 2.85 \text{ in}$$

Reduction Ratios:

The calculation of ratios can be found in Appendix A which outlines the gear ratio for each stage to achieve a speed reduction of 13.3. These speed ratios enable the gearbox to achieve the reduction while balancing load distribution and minimizing stress.

This table summarizes the characteristics of each gear in terms of the number of teeth, speed ratios, and pitch diameters. The table starts with the input gear (G2) and ends with the output gear (G9).

Gear	Teeth	Speed Ratios		Diameters	Pd and Pc Values	
G2	20	e3/2	-2.7	20	Pd1	1.00
G3	54	e4/3	1	54	Pd2	1.00
G4	20	e5/4	-2.7	20	Pd3	1.33
G5	54	e6/5	1	54	Pd4	1.33

G6	60	e7/6	-1.35	45	Pc1	3.14159265
G7	81	e8/7	1	60.75	Pc2	3.14159265
G8	60	e9/8	-1.35	45	Pc3	2.35619449
G9	81	e9/2	13.3	60.75	Pc4	2.35619449
All Gears have 25-degree pressure angle						

Table 3: Gear Dimensions and Speed Ratios

Center-to-Center Distance Sample Calculation:

The center-to-center distance between meshing gears is critical for the alignment and the engagement of the gear teeth.

For stages 1 and 2:

$$c = r_p + r_g = 10\ in + 27\ in = 37\ in$$

Line of Action Sample Calculation:

The line of action calculations determines the line along which the force acts.

For stages 1 and 2:

$$Z = \sqrt{\left(r_p + a\right)^2 - (r_p cos\emptyset)^2} + \sqrt{\left(r_g + a\right)^2 - (r_g cos\emptyset)^2} - csin\emptyset$$

$$Z = \sqrt{(10+1)^2 - (10\cos(25°))^2} + \sqrt{(27+1)^2 - (27\cos(25°))^2} - (37)\sin(25°)$$

$$Z = 4.21\ in$$

Contact Ratio Sample Calculation:

The contact ratio indicates the number of teeth that are in contact.

For stages 1 and 2:

$$c = \frac{Z}{P_b} = \frac{4.21\ in}{2.85\ in} = 1.48$$

Stage	Center to Center Distance [in]	Line of Action [in]	Contact Ratios
1	37	4.20	1.48
2	37	4.20	1.48
3	52.875	3.35	1.57

4	52.875	3.35	1.57

Table 4: Center-to-Center Distances, Line of Action, and Contact Ratio

8.3 Gearbox Power

The following calculations show the maximum and minimum power output for the output and input. The assumed efficiency is 96% for both the gearbox and the generator and is reflected in these calculations.

Gearbox output power from generator = 15 MW (max) 1 MW (min)

Gearbox efficiency = 96%
Generator efficiency = 96%

$$maximum\ gearbox\ ouput\ power\ P_o = \frac{15\ MW}{0.96} = 15.625\ MW$$

$$maximum\ gearbox\ input\ power\ P_i = \frac{15.625\ MW}{0.96} = 16.276\ MW$$

$$minimum\ gearbox\ ouput\ power\ P_o = \frac{1\ MW}{0.96} = 1.042\ MW$$

$$minimum\ gearbox\ input\ power\ P_i = \frac{1.042\ MW}{0.96} = 1.085\ MW$$

These power values are useful for determining the torque needed for other calculations.

8.4 Gearbox Torques

The following torque calculations were completed using the expected power and speed input and outputs.

$$maximum\ gearbox\ input\ torque\ T_{i,max} = \frac{P_{i,max}}{w_i} = \frac{16.276\ MW}{628.32\ rad/s} = 25.90\ kNm$$

$$maximum\ gearbox\ output\ torque\ T_{o,max} = \frac{P_{o,max}}{w_o} = \frac{15.625\ MW}{47.12\ rad/s} = 331.60\ kNm$$

$$minimum\ gearbox\ input\ torque\ T_{i,min} = \frac{P_{i,min}}{w_i} = \frac{1.085\ MW}{628.32\frac{rad}{s}} = 1.73\ kNm$$

$$minimum\ gearbox\ output\ torque\ T_{o,min} = \frac{P_{i,min}}{w_i} = \frac{1.042\ MW}{47.12\frac{rad}{s}} = 22.11\ kNm$$

The maximum torques are used to complete stress calculation which will help to determine the necessary dimensions to obtain reasonable safety factors. The torques seen by each gear are shown in the table below.

Torque	[kN·m]	[lbf·in]
Ti,max	25.9	229234.425
Ti,min	1.73	15311.7975
To,max	331.6	2934908.7
To,min	22.1	195601.575
T9max	331.6	2934908.7
T8max = Tp4	170.928	1512840.996
T7max	170.928	1512840.996
T6max = Tp3	88.170	780370.628
T5max	88.170	780370.628
T4max = Tp2	45.416	401965.662
T3max	45.416	401965.662
T2max = Tp1	23.410	207196.058

Table 5: Gear Torques

Table 5 also converts the value from kN·m to lbf·in for each gear, as imperial units will be used for all future calculations.

9. Sizing and Specifications

9.1 Gears

All eight gears will be spur gears made of ANSI 4340 steel with a 25-degree pressure angle. Earlier a breakdown was done on the specific gear diameters, contact ratios, and Pd and Pc values. ANSI 4340 steel was chosen because of its high strength and toughness, good fatigue resistance, and impact resistance with varying forces. However, 4340 steel is susceptible to corrosion if exposed to harsh environments. Additional precautions will need to be taken to protect gears. The 25-degree pressure angle was chosen to increase the load capacity, reduce wear over time and improve the contact ratio of the gears. The 25-degree pressure angle may add vibration, but strength is being prioritized. Below is a table outlining all the gear specs determined by the Pd values.

Addendum sample calculation:

For coarse pitch gears (Pd < 20):

$$addendum\ a = \frac{1}{Pd}$$

$$dedendum\ b = \frac{1.25}{Pd}$$

For stage 1:

$$a = \frac{1}{Pd} = \frac{1}{1} = 1\ in$$

$$b = \frac{1.25}{1} = 1.25\ in$$

The diameter of the addendum circle is given as $d_a = d + 2a = 20 + 2(1) = 22\ in$

The diameter of the dedendum circle is given as $d_b = d - 2b = 20 - 2(1.25) = 17.5\ in$

Full Depth Gear Tooth Specs	Gears 2, 4	Gears 3, 5	Gears 6, 8	Gears 7,9
Pressure Angle (˚)	25	25	25	25
Addendum [in]	1	1	0.83	0.83

Dedendum [in]	1.25	1.25	1.04	1.04
Addendum circle diameter [in]	22	56	51.67	69.17
Dedendum circle diameter [in]	17.5	51.5	47.92	65.42

Table 6: Gear Tooth Specifications

A drawing of gears 2 and 4 is shown below. The rest of the drawings can be found in Appendix C.

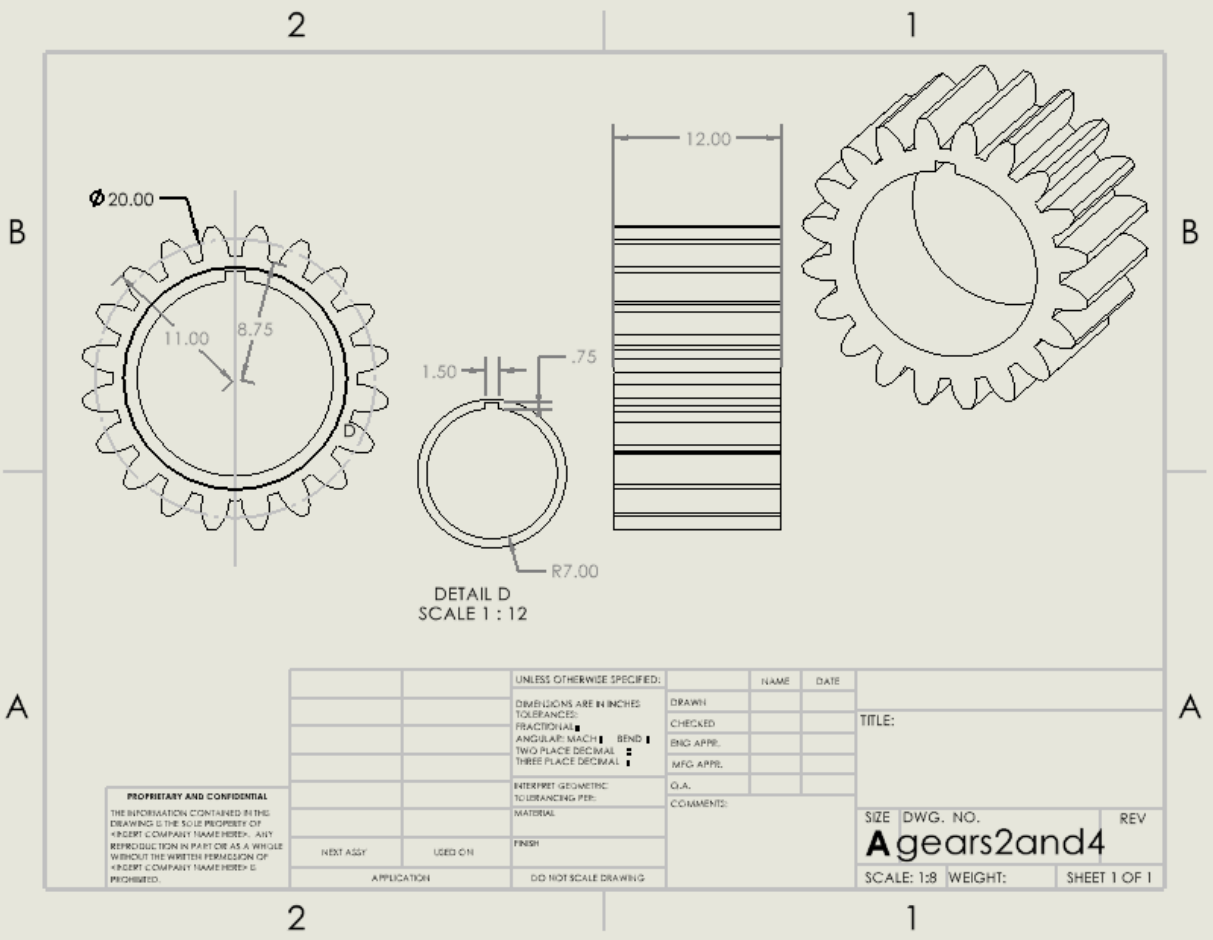


Figure 13: Drawing of Gears 2 and 4

9.1.1 Gears Critical Locations

The critical locations identified and analyzed for the gears included the tooth root fillet, contact point, and shaft connection. The tooth root fillet experiences high bending stress and therefore can experience fatigue failure if not correctly accounted for. At the contact point on the gear

teeth surface stress is at its highest. Therefore, surface stress calculation was completed. If gears are not correctly meshed, issues such as pitting can arise. Lastly, connection to the shafts holds potential for stress raisers. To help reduce stress sharp edges and corners were avoided.

9.1.2 Gears Failure Analysis

Below is a sample calculation of one of the bending stresses experienced at the tooth root fillet. A total value of 29815.93 kpsi was used to ensure appropriate safety factors.

Bending Stress Sample Calculation:

$$\sigma B = \left(\frac{\mathrm{Wtmax} * \mathrm{Pd}}{\mathrm{F} * \mathrm{J}}\right) * \left(\frac{(\mathrm{Ka} * \mathrm{Km})}{\mathrm{Kv}}\right) * \mathrm{KS} * \mathrm{KB} * \mathrm{KI}$$

$$\sigma B = \left(\frac{130440 * 1.33}{13.33 * 0.4}\right) * \left(\frac{(1.75 * 1.75)}{0.89}\right) * 1 * 1 * 1$$

$$\sigma B = 112041.9 \text{ [psi]}$$

Below is a sample calculation of one of the surface stresses experienced at the contact point. A total value of 56076.0 kpsi was used to ensure appropriate safety factors.

Surface Stress Sample Calculation:

$$\sigma c = Cp * sqrt\left(\frac{\mathrm{Wtmax}}{\mathrm{F} * \mathrm{I} * \mathrm{d}} * \left(\frac{\mathrm{Ca} * \mathrm{Cm}}{\mathrm{Cv}}\right) * \mathrm{Cs} * \mathrm{Cf}\right)$$

$$\sigma c = 2300 * \mathrm{sqrt}\left(\frac{30569}{12 * 0.739 * 15} * \left(\frac{1.5 * 1.5}{0.87}\right) * 1 * 1\right)$$

$$\sigma c = 56076.0 \text{ [psi]}$$

To ensure the gears were able to withstand the high contact forces both bending and surface stress calculations were completed. Below is a Safety factor sample calculation.

Safety Factors Sample Calculation:

$$NB = \left(\frac{SB}{\sigma B}\right)^{p} = \frac{29894}{141000} = 4.717$$

The following table shows the values for the bending fatigue safety factor (NB) and Surface fatigue safety factor (NC).

Safety Factor NB		Safety Factor NC	
NB1	8.3	NC1	11.6
NB2	2.5	NC2	3.5
NB3	1.7	NC3	1.4
NB4	1.3	NC4	1.3

Table 7: Safety Factors for Stress Calculations on Gears

The Static Safety factors all are acceptable. Following this, cyclic loading conditions were then checked to ensure an appropriate lifetime. The endurance limit marks the threshold below which the steel gears can theoretically withstand an infinite amount of cyclic loading without failure. Two SN diagrams were made for both the axial and bending conditions experienced by the gears. The ultimate strength of 250 kpsi was used as well as relevant correction factors shown in Appendix A, Figure A.2. There are two graphs for the two different loading conditions.

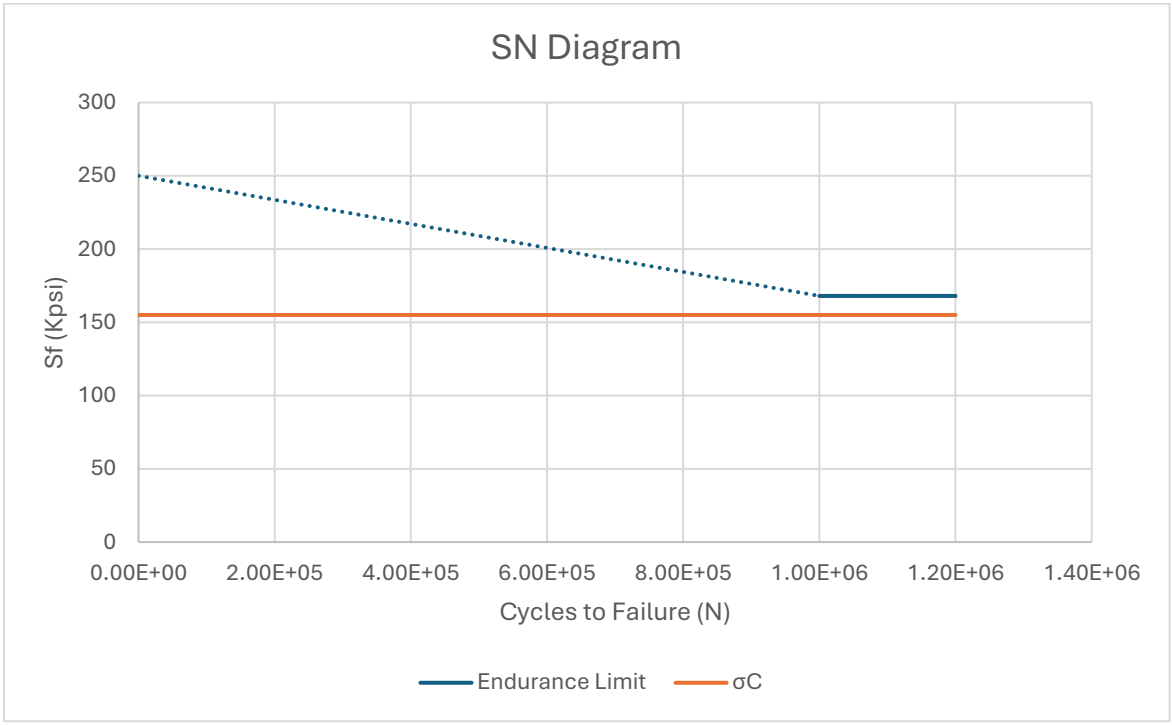


Figure 14: S-N Diagram for Gears Surface Stress

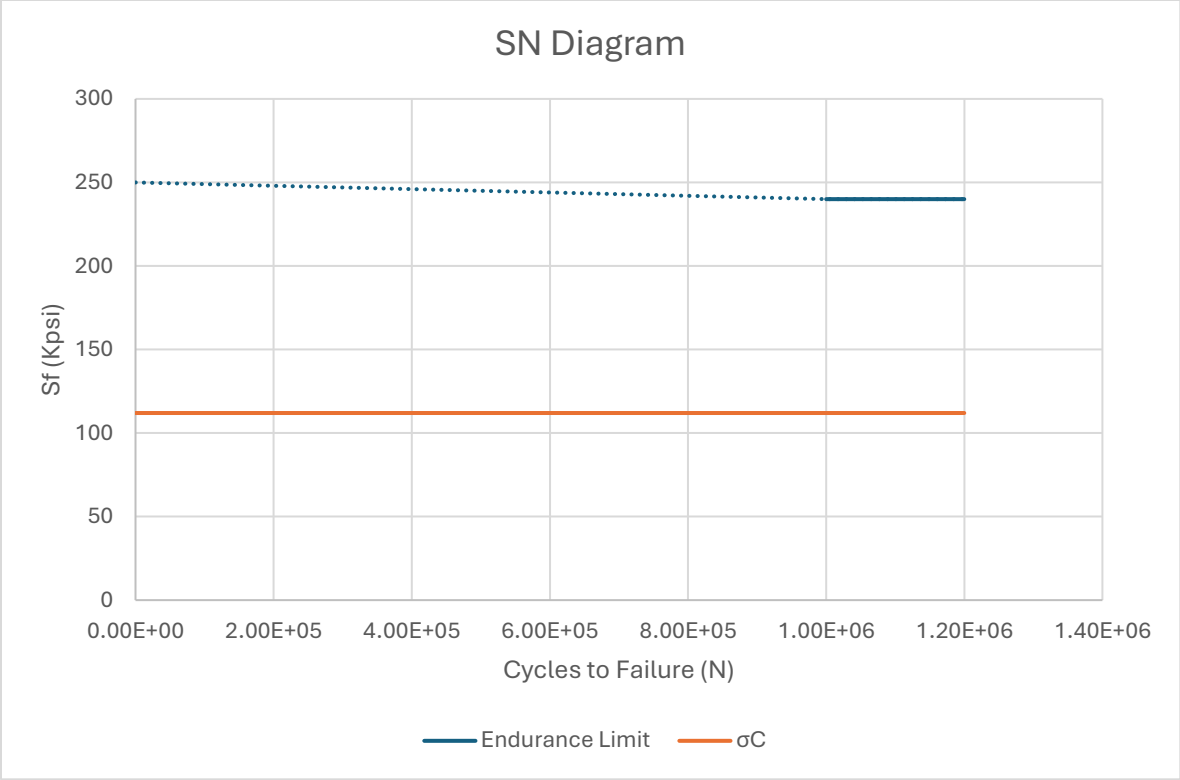


Figure 15: S-N Diagram for Gears Bending Stress

Since the highest bending stress of 112.0 kpsi is below the endurance limit of 133.33 kpsi, and the highest surface stress of 155 is below the endurance limit of 165 the failure due to the number of cycles can be considered negligible. To increase the endurance limit to the values shown above the following factors are considered. Extra emphasis will be put towards improving the surface finish, high quality lubrication, increase material hardness, and a surface hardening coating if necessary.

9.2 Bearings

The bearings chosen are Double Row Tapered Roller Bearings made up of through-hardened steel from NTN Bearing Corporation [9]. This bearing has an inner diameter of 14.1 inches, an outer diameter of 23.2 inches, and a width of 7.6 inches. The larger width increases stability, leading to less vibration and maintenance over the 25-year life. A drawing of the bearing is shown below:

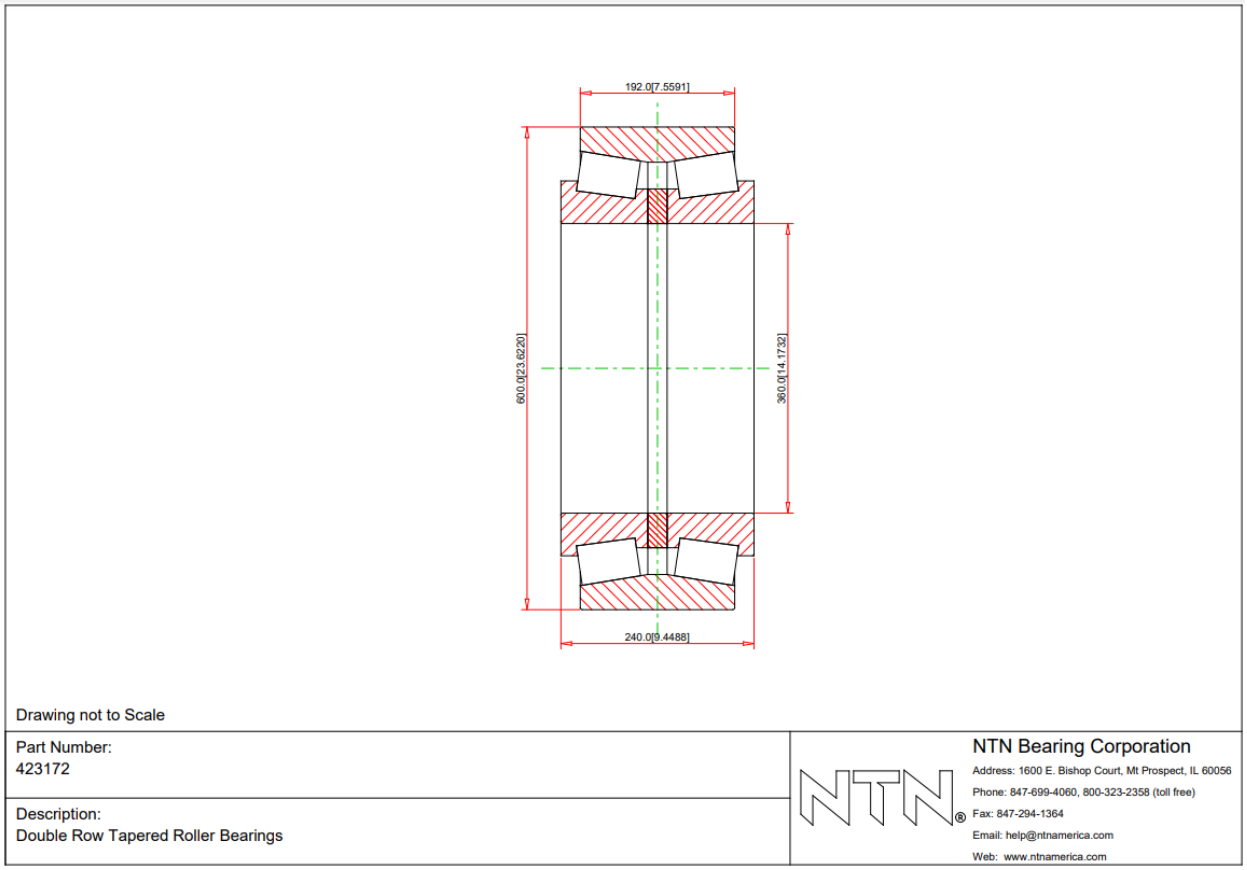


Figure 16: Double Row Tapered Roller Bearing [9]

9.1.1 Bearings Critical Locations

The material can handle large loads. Since we are using spur gear the axial forces are negligible and therefore axial strength was not considered during the decision process. Based on the reaction forces on the shaft, discussed in 9.5 Shafts, the highest reaction forces the bearing must withstand is 12882.28 lbf. The static load rating for the provided bearing is 140,000 lbf, which is well above. Additionally, the bearing has a dynamic load rating of 720,000 lbf ensuring high reliability during the high dynamic loads. Surface fatigue needs to be analyzed using the hour of operation calculations.

9.1.2 Bearings Failure Analysis

Since there is no radial load the P is determined by the radial load (12882.28 lbf) and the ratio factor for the inner ring (1.0).

Basic Life Rating Sample Calculation:

$$L10 = \left(\frac{C}{P}\right)^{p} = \left(\frac{720000\ [lbf]}{12882.28\ [\text{lbf}]}\right)^{\frac{10}{3}} = 667522.4$$

Based on the above basic life calculation the bearing is expected to achieve a L10 of 52485.7 million revolutions.

Hours of Operation Sample Calculation:

$$L10h = L10 * \left(\frac{10^6}{60n}\right) = \ 5006919\ hours$$

Based on the estimated hours of operation the expected lifetime of the bearing is 571 years. This is above the target lifetime of 25 years. The table below shows the predicted lifetime of all the bearings at different stages in the gearbox.

Stage	W_r	L_{10}	Hours	Years
1	12882	667522	5006919	572
2	24992	73301	1484418	169
3	14556	444287	12158924	1388
4	20902	132985	4925359	562

Table 8: Lifetime of Bearings

9.3 Couplings

Couplings will be used in two different applications such as connecting the output shaft to the generator and connecting the input shaft to the steam turbine. Before sourcing max torque, nominal torque, RPM, and shaft diameter were considered. The torque calculations are shown below:

$$Output\ Torque = \frac{15.625[MW]}{47.12[rad/s]} * \left(\frac{8850.75lbin}{1kN}\right) = 2934908.7lbfin$$

$$Input\ Torque = \frac{16.275[MW]}{628.32[rad/s]} * \left(\frac{8850.75lbin}{1kN}\right) = 229269.81lbfin$$

Due to the differing torque and speed requirements of the output and input shafts, two distinct couplings were selected.

Output Shaft Coupling

For the output shaft, the size 1200 Standard Grid Style Couplings with a Horizontal Cover was chosen. The coupling is designed to connect the 12 in stepped down part of the output shaft. Nominal torque can be assumed to be about half the peak torque value, making the minimum nominal torque rating about 1,500,000 in-lb. The coupling features a nominal torque rating of 1,650,000 in-lbs and a maximum speed of 900 RPM. These values exceed the expected operating conditions. The coupling design ensures high performance for high torque and lower speed operations. A drawing of the output shaft coupling is shown below:

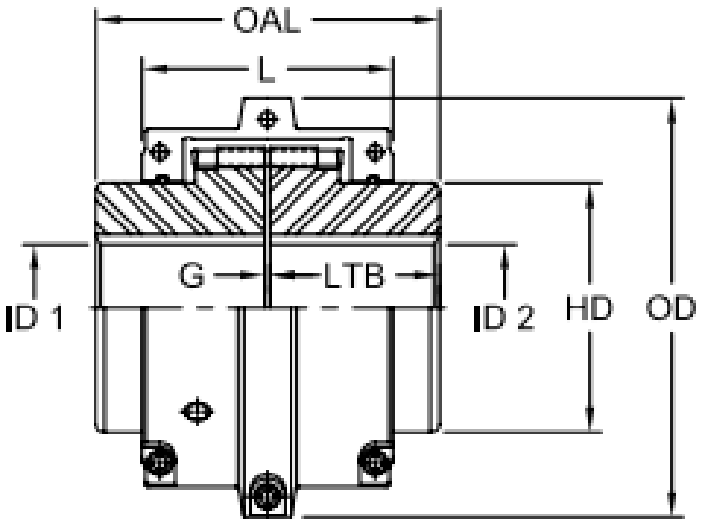


Figure 17: Standard Grid Style Couplings with a Horizontal Cover [10]

Input Shaft Coupling

For the input shaft, a size 6 FHS Type High-Speed Close Coupled Couplings was selected. This coupling can accommodate a maximum torque of 731,000 in-lbs and supports a maximum rotational speed of 9000 RPM. This exceeds the expected operating conditions. The coupling is made from AISI 1045 steel and is precision ground to ensure dynamic stability. To use this coupling the input shaft must have a diameter of 4.5 inches. A drawing of the input shaft coupling is shown below:

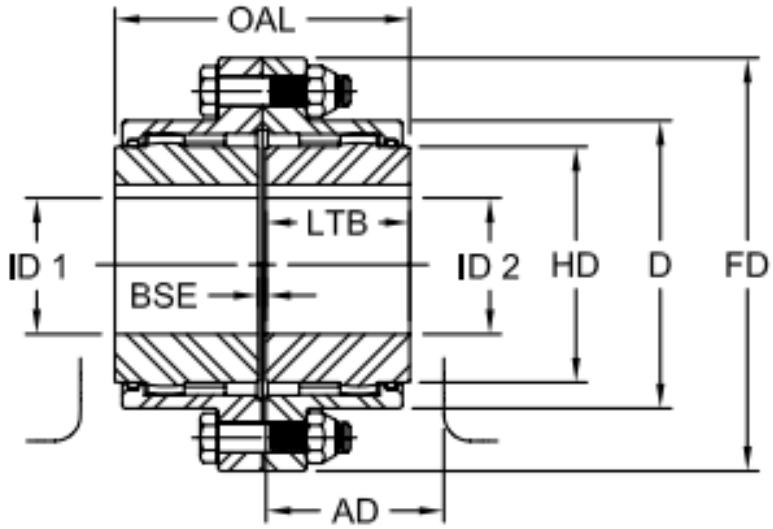


Figure 18: FHS Type High-Speed Close Coupled Couplings

9.4 Seals

Rotary Shaft Seals were chosen for the input and output shafts. The seals are designed to keep out contaminants, preventing shorter component life. The SKF [12] seals are chosen to have a bore diameter of 15” and a width of 0.5” to fit the 14” shaft diameter. The seal should be inspected during routine maintenance.

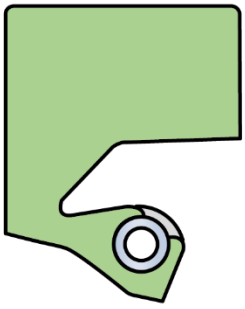


Figure 19: SKF Rotary Shaft Seals

9.5 Housing

To select a material for the housing, the design team considered factors such as strength and load-bearing capacity, vibration and noise damping, thermal conductivity and heat dissipation, weight, and cost. The housing will include ports for the lubrication circulation system and will be

securely mounted to the steel supporting frame. The housing will also have holes to accommodate the input and output shafts.

Factor	Cast Iron	Steels	Aluminum
Strength and Load-Bearing Capacity	High strength and good load-bearing capacity; suitable for most industrial applications	Very high strength and load-bearing capacity; ideal for high-stress applications	Moderate strength; suitable for medium-load applications
Vibration and Noise Damping	Excellent damping properties	Moderate damping properties	Poor damping properties
Thermal Conductivity and Heat Dissipation	Moderate thermal conductivity	Moderate thermal conductivity	High thermal conductivity
Weight	Heavy	Very heavy	Light
Cost	Generally low cost	Moderate-high cost	Moderate cost

Table 9: Material Options for Housing

Based on the table above, the design team decided to build the housing with cast iron due to its high strength and load-bearing capacity, strong damping abilities, and relatively low cost compared to other options.

The two-piece clamshell-style housing is designed to follow the natural form of the gear train, with the locating features post-machined into the cast iron structure. Included is a forced lubrication system circulating oil from the wells to the tops of certain gears. This design increases rigidity and reduces vibration.

The recommended circulating Lubricant is Valvoline Epg S320 Extreme Pressure Gear Oil. This is due to its ability to withstand high loads and large shocks. [16]

To attach the two halves of the housing together securely, 37 SAE grade 5, 1”-8 UNC hex head bolts shall be used with corresponding nylon locking nuts. The bolts shall be lubricated using assembly grease and tightened to a torque of 720 lb.-ft.

To mount the base of the housing to the foundation, 28 1”-8 9-inch-long concrete stud anchors shall be used. They shall be torqued to 300 ft-lb upon installation.

9.6 Shafts

9.6.1 Shaft Loads

To begin shaft analysis, three-dimensional free-body diagrams were developed for each input, intermediate, and output shaft. The following naming scheme was used when creating the free-body diagrams:

- R_{XY} = reaction force of bearing X in y-direction
- R_{XZ} = reaction force of bearing X in z-direction
- W_{rX} = radial force of gear X
- W_{tX} = tangential force of gear X
- T = torque of the given shaft
- C_{XY} = contact point between gear X and gear Y

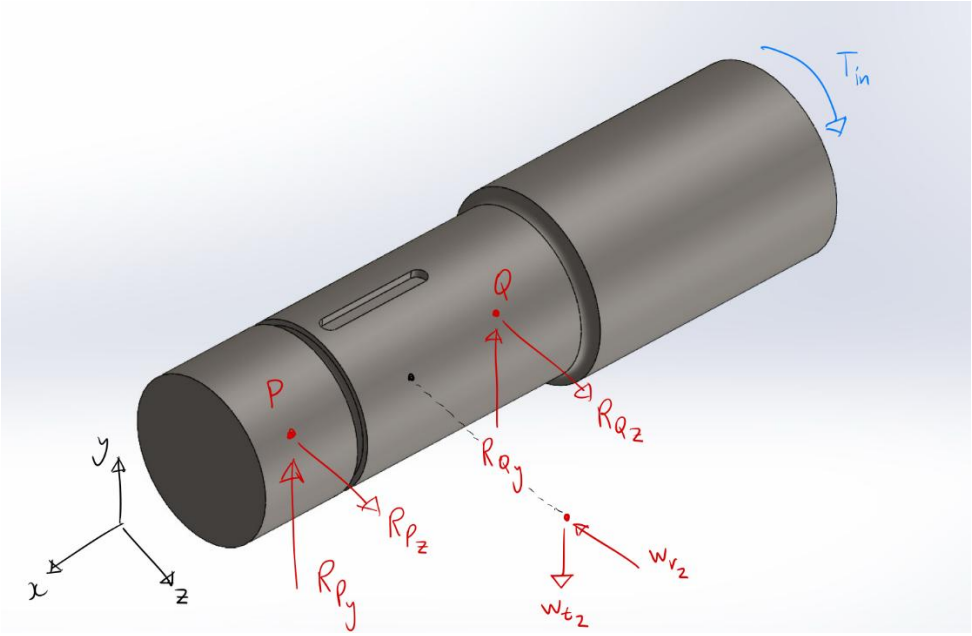


Figure 20: Shaft 1-2 Free Body Diagram

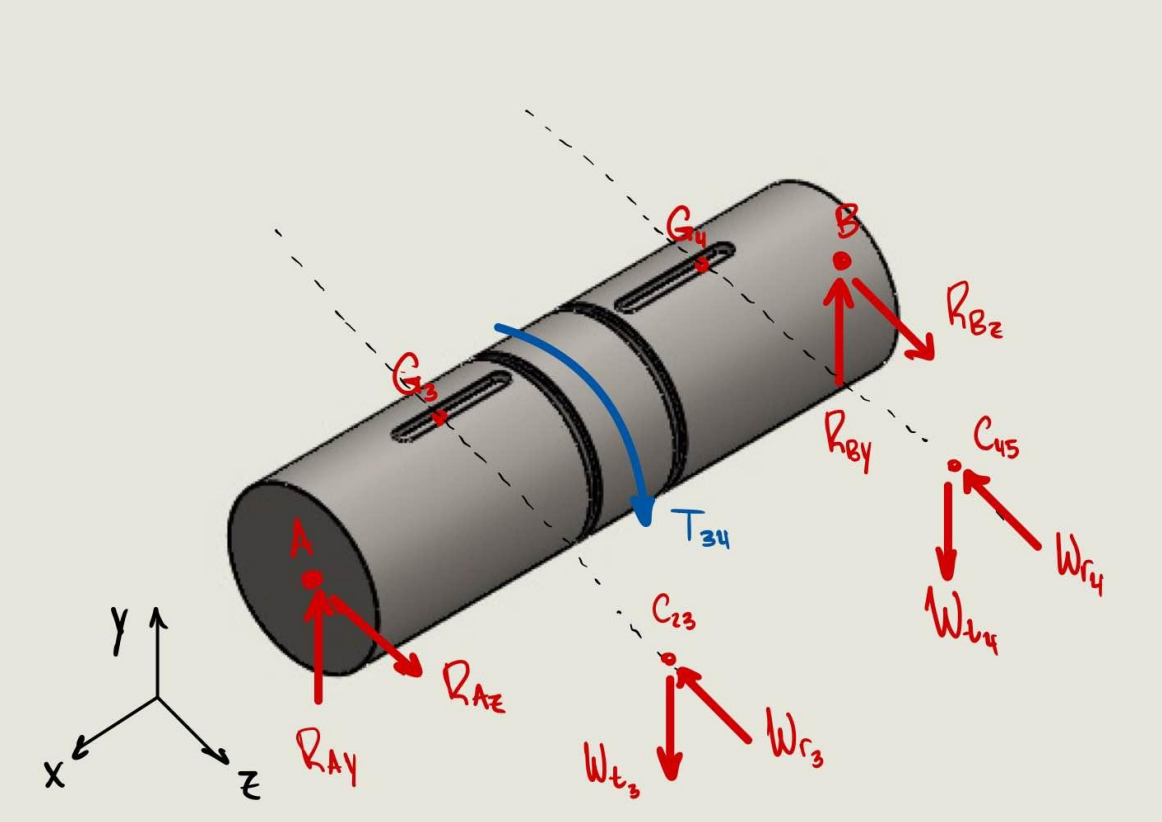


Figure 21: Shaft 3-4 Free Body Diagram

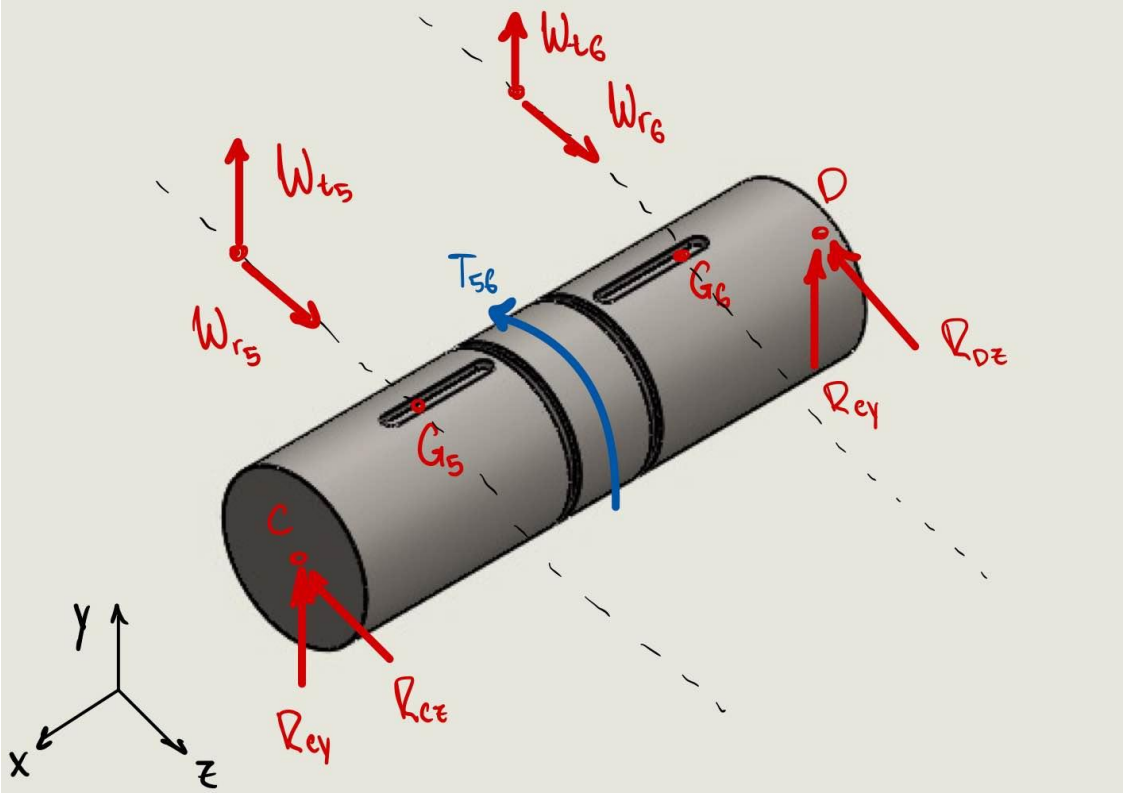


Figure 22: Shaft 5-6 Free Body Diagram

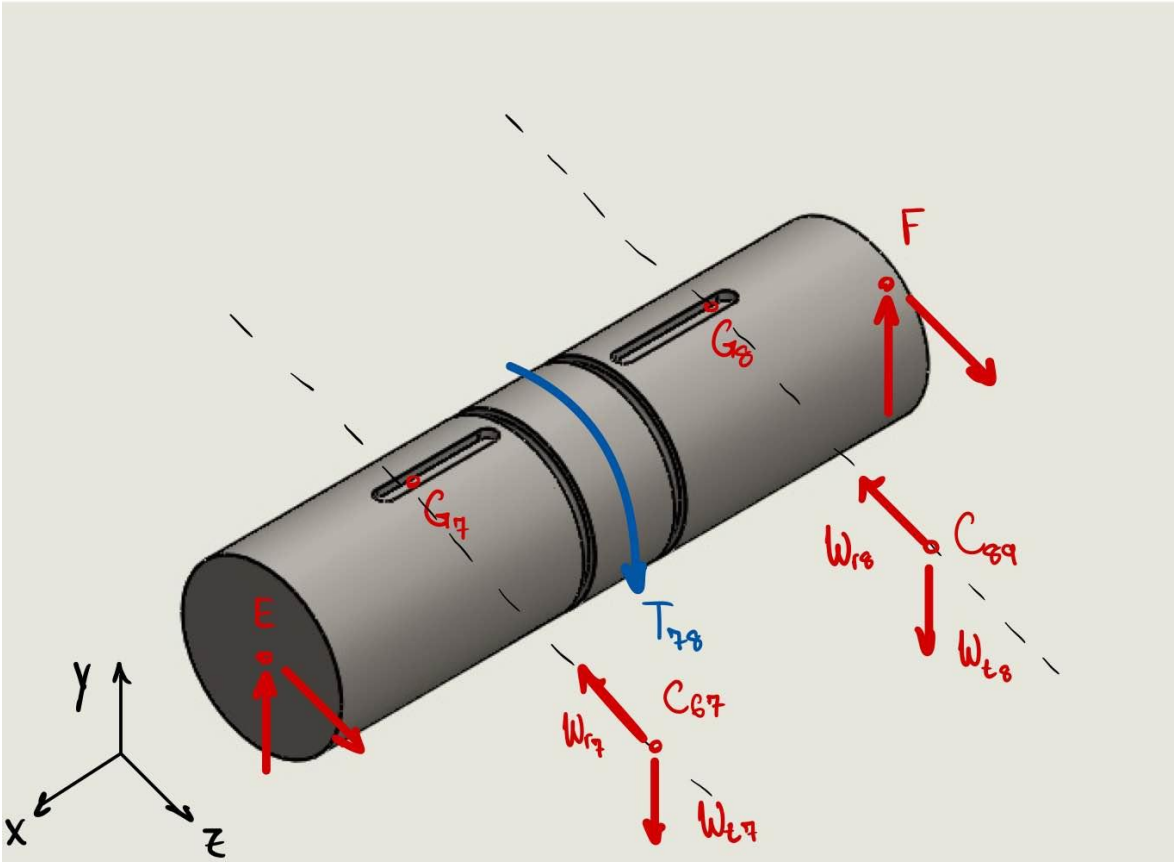


Figure 23: Shaft 7-8 Free Body Diagram

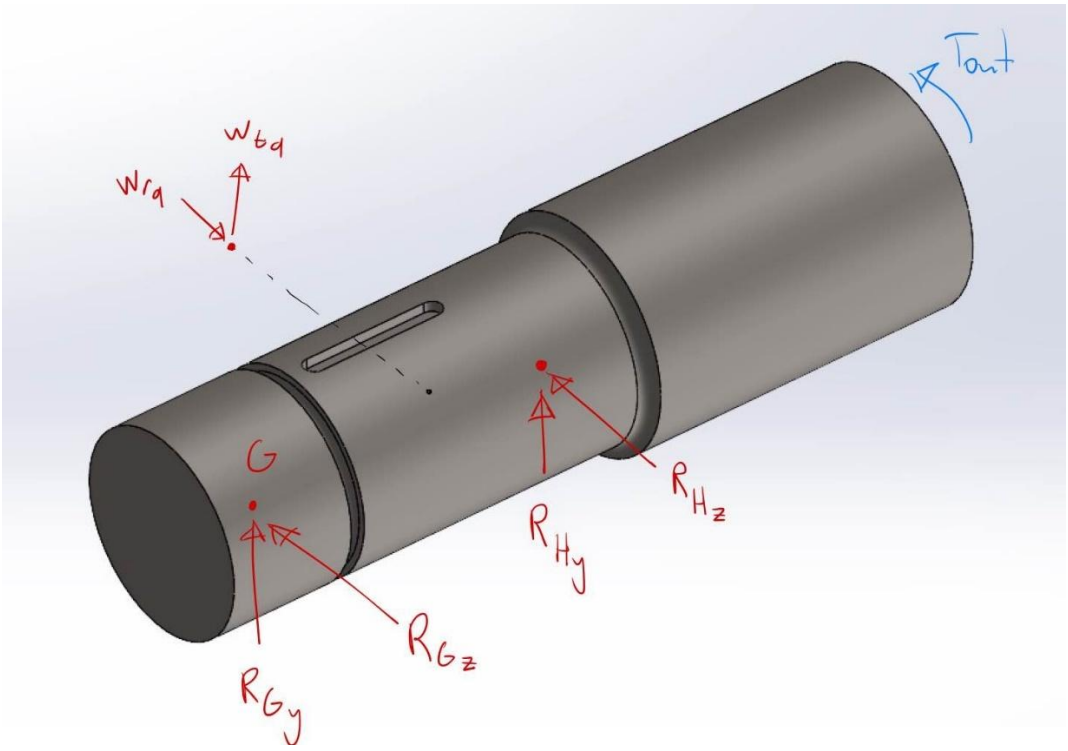


Figure 24: Shaft 9-10 Free Body Diagram

The torques for each shaft are detailed in Table 5. These torque values were used to determine the radial and tangential forces for each gear using the following equations, with the results summarized in Table 10.

$$W_t = \frac{T_P}{r_P}$$

$$W_r = W_t * tan\emptyset$$

	Gear 2	Gear 3	Gear 4	Gear 5	Gear 6	Gear 7	Gear 8	Gear 9
Wt [lbf]	27626.14	27626.14	53595.42	53595.42	31214.83	31214.83	44824.92	44824.92
Wr [lbf]	12882.28	12882.28	24991.96	24991.96	14555.71	14555.71	20902.2	20902.2
Thickness [in]	12	12	12	12	13.33	13.33	13.33	13.33

Table 10: Gear Loads

The geometric parameters of each shaft (input, intermediate, and output) are described by the drawings found in Appendix B. In correlation with the geometric parameters, a static analysis of each shaft was completed to calculate the reaction forces at each of the bearings. This was done by summing the moments around one end to find the reaction force at the opposite end, followed by summing the forces in each direction. The results are summarized in Table 11.

Reaction Forces					
Shaft 1-2		**Shaft 3-4**		**Shaft 5-6**	
RPy [lbf]	14140.39	RBy [lbf]	50657.6	RDy [lbf]	36809.97
RPz [lbf]	6593.774	RBz [lbf]	23622.03	RDz [lbf]	17164.77
RQy [lbf]	13485.75	RAy [lbf]	30563.96	RCy [lbf]	48000.27
RQz [lbf]	6288.507	RAz [lbf]	14252.21	RCz [lbf]	22382.89
Shaft 7-8		**Shaft 9-10**			
RFy [lbf]	45751.25	RHy [lbf]	23349.04		
RFz [lbf]	21334.16	RHz [lbf]	10887.84		
REy [lbf]	30288.5	RGy [lbf]	21475.88		
REz [lbf]	14123.76	RGz [lbf]	10014.36		

Table 11: Reaction Forces at Each Bearing

9.6.2 Fatigue Analysis

Once the loads of each shaft have been determined, we can begin fatigue analysis. To begin, we must first calculate each shaft's loading type. As proven by the free body diagrams, we have multiaxial loading conditions (bending stress and torsional shear in the x-direction and y-direction). The maximum bending stress was determined by creating force and moment diagrams of each shaft. The results of each moment diagram are summarized in Table 12.

	Shaft 1-2	Shaft 3-4	Shaft 5-6	Shaft 7-8	Shaft 9-10
Max Moment (x-dir) [lbf-in]	145646.1	425523.9	529994.4	384310.5	246220.9
Max Moment (y-dir) [lbf-in]	67915.87	198425	247140.5	179206.9	114814.7
Minimum Moment (x-dir) [lbf-in]	0	0	0	0	0
Minimum Moment (x-dir) [lbf-in]	0	0	0	0	0
Maximum Torsion [lbf-in]	207196.1	401965.7	780370.6	1512841	2934909

Table 12: Maximum and Minimum Loading Conditions

From using the values in Table 12, the associated values of maximum/minimum bending stress and maximum torsional shear stress can be calculated. Due to the multiaxial loading conditions, each stress value will need to be computed in its respective axis of application. Table 13 summarizes the results of each stress calculation.

$$Bending\ stress = \ \sigma = \frac{M * c}{I}$$

$$Torsional\ shear\ stress = \ \tau = \frac{T * c}{J}$$

Shaft Stresses					
	Shaft 1-2	Shaft 3-4	Shaft 5-6	Shaft 7-8	Shaft 9-10
Maximum Bending Stress (x-dir) [psi]	270.3241	789.7869	983.6878	713.2934	349.4141
Maximum Bending Stress (y-dir) [psi]	126.0542	368.2837	458.7012	332.6142	162.9345
Minimum Bending Stress (x-dir) [psi]	0	0	0	0	0
Minimum Bending Stress (x-dir) [psi]	0	0	0	0	0
Maximum Torsional Shear Stress [psi]	769.126	1492.124	2896.789	5615.77	10894.58

Table 13: Maximum and Minimum Shaft Stresses

Since the minimum stress is equal to zero, this means that our shaft is under multiaxial fluctuating (repeated) loading, which is Category 4 loading. The following dimensions and parameters were used for all shafts (input, intermediate, and output) in the fatigue life calculations.

- Major diameter: outside diameter of the shaft

- Input/Output diameter: applicable to shafts with shoulders (input and output shafts), represents the larger outer diameter of the shoulder
- First & second moment of inertia: constant value for all shafts (as all shafts have the same diameter)
- Groove radius: radius of the groove utilized to locate the clamping collars
- Keyway radius: radius of the keyway utilized to radially support the gears

Shaft Dimensions		Units
Major Diameter	14	in
Input Diameter (for shaft 1-2)	6	in
Output Diameter (for shaft 9-10)	12	in
Moment of Inertia	3771	in^4
Second Moment of Interia	1886	in^4
Groove Radius	0.5	in
Shoulder Radius (if applicable)	1	in
Keyway Radius	0.75	in

Table 14: General Shaft Dimensions

The shafts and gears for this project are made of 4340 Steel, quenched and tempered at 600°F, to ensure high strength and durability. To determine the corrected endurance limit (Se) for the shafts, several correction factors were applied. The load factor (C_{load}) and temperature factor (C_{temp}) were both assigned a value of 1.0, as the shafts experience purely torsional and bending stresses and are expected to operate below 450°F. The size factor (C_{size}) was set at 0.6 because the major diameters of the shafts, measuring 14”, exceed the 10” threshold requiring adjustment. The surface finish factor (C_{surf}) was calculated using Table 6-3 from Norton, with constants A=16.841 and b=−0.265. For reliability, a factor (C_{reliab}) of 0.814 was chosen to account for 99% reliability, ensuring proper performance. With these correction factors defined, the endurance limit (Se) was calculated and utilized to determine the factor of safety for each shaft. The parameters discussed above are summarized in Table 15.

$$Endurance\ limit = Se = C_{load}C_{size}C_{surf}C_{temp}C_{reliab}Se'$$

Correction Factors and Material Properties	
Material	4340 Steel Quench and Temper @ 600F
Yield Strength [kpsi]	230
Ultimate Strength [kpsi]	250
C-load	1
C-size	0.6
C-surf	0.625
C-temp	1
C-reliab	0.814
Notch Sensitivity (q)	0.97
Se' [kpsi]	100
Se [kpsi]	30.53

Table 15: Correction Factors and Material Properties

The geometry and features of the shafts, designed to locate and secure gears and bearings, result in localized stress concentrations at specific points. The input and output shafts are equipped with grooves and shoulders to axially constrain the gears and bearings, as well as keyways to radially constrain the gears. As a result, each of these shafts exhibits three critical locations where stress concentrations are significant. In contrast, the intermediate shafts, while featuring keyways and grooves, lack shoulders and therefore do not require stress analysis at those points.

Stress concentration factors were calculated using Appendix C in Norton, identifying values for parameters A and b for both torsion and bending in groove and shoulder configurations. These values were used to determine K_t, which was converted to K_f using the notch sensitivity factor q (as detailed in Norton, Figure 6-36). We can then multiply the found nominal stresses by K_f to compare those values to the yield strength S_y.

Throughout the design process, the calculated stresses consistently remained below S_y, confirming that the design is structurally sound and concluding that $K_{fm} = K_f$

In order to determine the safety factor for each critical location, we had to combine the multiaxial loading into a pair of Von Mises stresses by the equation:

$$Von\ Mises\ Stress = \sigma' = \sqrt{\sigma_x^2 + \sigma_y^2 - \sigma_x\sigma_y + 3\tau_{xy}^2}$$

This equation uses the nominal stresses multiplied by their respective stress concentrations. Finally, the mean and amplitude stresses can be combined with the material properties to find the safety factor using the following equation. The results are summarized in Tables 16-18 below.

$$Factor\ of\ Safety = N = \frac{S_e S_{ut}}{\sigma'_a S_{ut} + \sigma'_m S_e}$$

Groove					
	Shaft 1-2	**Shaft 3-4**	**Shaft 5-6**	**Shaft 7-8**	**Shaft 9-10**
Kt Torsion	1.63	1.63	1.63	1.63	1.63
Kf Torsion	1.61	1.61	1.61	1.61	1.61
Kt Bending	2.29	2.29	2.29	2.29	2.29
Kf Bending	2.25	2.25	2.25	2.25	2.25
σa Torsion [psi]	612.50	1188.20	2306.80	4472.00	8675.71
σm Torsion [psi]	612.50	1188.20	2306.80	4472.00	8675.71
σa Bending [psi]	438.95	1419.60	1446.60	1248.40	567.38
σm Bending [psi]	438.95	1419.60	1446.60	1248.40	567.38
σa' [psi]	1092.11	2222.50	4086.00	7780.80	15030.51
σm' [psi]	1092.11	2222.50	4086.00	7780.80	15030.51
N	24.91	12.40	6.63	3.50	1.81

Table 16: Stress Concentration Factors, Stresses, and Factor of Safety for Shaft Grooves

Shoulder					
	Shaft 1-2	**Shaft 3-4**	**Shaft 5-6**	**Shaft 7-8**	**Shaft 9-10**
Kt Torsion	1.72	N/A	N/A	N/A	1.93
Kf Torsion	1.70	N/A	N/A	N/A	1.87
Kt Bending	1.35	N/A	N/A	N/A	1.62
Kf Bending	1.34	N/A	N/A	N/A	1.58
σa Torsion [psi]	654.3	N/A	N/A	N/A	9267.51
σm Torsion [psi]	654.3	N/A	N/A	N/A	9267.51
σa Bending [psi]	265.9	N/A	N/A	N/A	343.69
σm Bending [psi]	265.9	N/A	N/A	N/A	343.69
σa' [psi]	1144.1	N/A	N/A	N/A	16053.09
σm' [psi]	1144.1	N/A	N/A	N/A	16053.09
N	23.78	N/A	N/A	N/A	1.69

Table 17: Stress Concentration Factors, Stresses, and Factor of Safety for Shaft Shoulders

Keyway					
	Shaft 1-2	Shaft 3-4	Shaft 5-6	Shaft 7-8	Shaft 9-10
Kt Torsion	2.50	2.50	2.50	2.50	2.50
Kf Torsion	2.46	2.46	2.46	2.46	2.46
Kt Bending	2.50	2.50	2.50	2.50	2.50
Kf Bending	2.46	2.46	2.46	2.46	2.46
σa Torsion [psi]	944.1	1831.6	3555.8	6893.4	13373.10
σm Torsion [psi]	944.1	1831.6	3555.8	6893.4	13373.10
σa Bending [psi]	486.6	1573.6	1603.5	1383.8	628.9
σm Bending [psi]	486.6	1573.6	1603.5	1383.8	628.9
σa' [psi]	1660.3	3305.9	6231.3	11967.6	23165.9
σm' [psi]	1660.3	3305.9	6231.3	11967.6	23165.9
N	16.39	8.29	4.35	2.27	1.17

Table 18: Stress Concentration Factors, Stresses, and Factor of Safety for Shaft Keyways

An analysis of the shafts reveals a decreasing trend in the safety factor from the input shaft to the output shaft. This aligns with expectations, as both torsional and bending forces increase with the gear ratios. Additionally, safety factors vary across different areas of stress concentrations, with shoulders exhibiting the highest safety factors, followed by grooves, and then keyways.

For the final (output) shaft, designated as 9-10, the safety factors are calculated as 1.81, 1.69, and 1.17 for the shoulders, grooves, and keyways, respectively. While the safety factors for the shoulders and grooves exceed the design minimum of 1.5, the keyway safety factor, at 1.17, falls below this threshold but remains above the critical limit of 1.

9.6.3 Cyclic Loading Conditions

The calculated safety factors can be further validated using an S-N curve. The following S-N curve represents the material's ultimate strength at 0 cycles and its calculated endurance limit at $1x10^6$ cycles. This curve represents the material's strength over cyclic loading conditions.

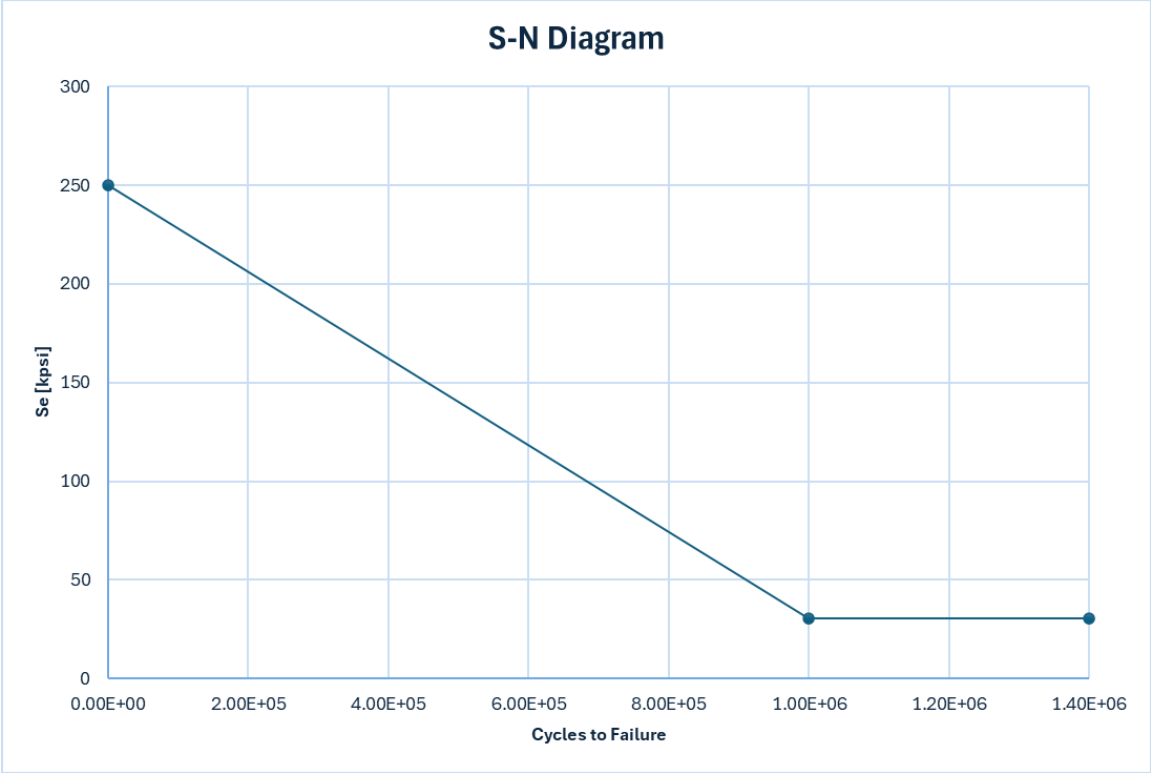


Figure 25: S-N Diagram for Shafts

From analyzing the S-N curve, we can determine that the stresses displayed in Tables 16, 17, and 18, are significantly below the material's calculated endurance limit. Thus, fatigue failure due to cyclic loading can be neglected. Since the maximum stress from each table is approximately 50% of the material's endurance limit, the impact of cyclic loading conditions and variations in the energy cycle (such as start/stop operations) can be considered negligible.

9.6.4 Misalignment Effects

Shaft misalignment occurs when the shaft and bearing axes are not colinear, leading to uneven loading conditions. It can be caused by manufacturing tolerances, assembly errors, wear, or thermal expansion during operation. Misalignment leads to uneven stress distribution, increased vibration, heat generation, premature bearing wear, and reduced life. The effect of shaft misalignment can be quantified using deflection analysis (how much the shaft deflects due to misalignment). For a shaft under load, the deflection equations were chosen for a simply supported beam under torsional and center moment loading conditions [14].

$$Bending\ Deflection\ [in] = \delta = -\frac{Mx}{24LEI}(L^2 - 4x^2) \rightarrow (0 \leq x\ \leq \frac{L}{2})$$

$$Angular\ Misalignment\ [deg] = \ \emptyset = \ tan^{-1}(\frac{\delta}{L})$$

$$Torsional\ Deflection\ [deg] = \ \theta = \frac{TL}{GJ} * \frac{180}{\pi}$$

From using the parameters of each shaft (dimensional drawings can be found in Appendix B for each shaft) outlined in Tables 12 and 14, along with the values below, the deflections of each beam can be quantified. The Young's Modulus and Shear Modulus for 4340 Steel were sourced from the appendix of Norton: E = 30 x 10^6 psi, G = 11.7 x 10^6 psi.

Shaft Deflection					
	Shaft 1-2	**Shaft 3-4**	**Shaft 5-6**	**Shaft 7-8**	**Shaft 9-10**
Bearing to Moment Distance (x) [in]	10.3	12	12	13.3	11.5
Bearing to Bearing Distance (L) [in]	21.1	40.4	40.4	41.7	23.93
Bending Deflection [in]	-5.45902E-07	-4.9E-05	-6.1E-05	-4.7E-05	-1.902E-06
Angular Misaslignment [deg]	-2.58722E-08	-1.2E-06	-1.5E-06	-1.1E-06	-7.947E-08
Torsional Deflection [deg]	0.01135321	0.042172	0.081872	0.163826	0.1823862

Table 19: Bending and Torsion Deflection of Shafts

The deflections and misalignments of each shaft, as shown in the table, are negligible. This outcome aligns with expectations, given the high strength of the selected material and the relatively low applied stress levels in comparison to the material's endurance limit. Although the bearing misalignment tolerance for the selected bearings is not listed in the technical specifications, we can assume a tolerance misalignment of 0.05 rad (2.86 deg). The calculated misalignment is significantly lower than this threshold, indicating safe operation under current loading.

To ensure long-term reliability, despite the small deflections calculated, the following measures should be implemented. Use precise alignment tools, such as laser alignment systems, during installation to minimize initial misalignment. Monitor shaft and bearing alignment using vibration analysis or thermal imaging techniques to detect early signs of misalignment. Considering the long-term reliability goal of 25 years, proactive measures, such as the ones listed

above, should be taken to ensure reliable operation and minimize future potential of misalignment

9.6.5 Fasteners

The only fasteners in the shaft system are those used to secure the locking collars to the grooves in the shafts. These collars serve to constrain the gears and bearings axially along the shafts and are designed to split for easy installation and removal while maintaining their function. The split gap in their design, as shown in the drawings in Appendix B, allows the bolts and the collars to be placed under slight tension when tightened. This tension prevents any looseness or potential slippage of the fasteners.

The collars are tightly dimensioned to fit the grooves in the shafts and experience minimal radial mechanical load. Their primary load is axial, resulting from the movement of the shafts, bearings, spacers, and gears. This axial force does not affect the fasteners, as the bolts are only responsible for securing the collars radially within the shaft grooves. Since the bolts are not subjected to fluctuating or reversed loading, fatigue calculations are unnecessary. As the two split halves of the collars do not physically touch, there is no preload applied to the bolts, keeping stresses low and ensuring a high factor of safety.

10. Assembly and Setup Process

The following steps outline how to assemble the gearbox.

1. Prepare a foundation, to support the ~60 000 kg mass of the speed reducer, a robust foundation should be installed in the area of operation.
2. Prepare Housing: Attach the oil sump ducts to the holes in the housing base and install reservoir, pumps, and oil filters. Ensure housing is free from debris.
3. Ensure bearings are adequately lubricated.
4. Assemble the Input shaft.
    a. Onto the input shaft, slide a bearing and a 2-inch spacer
    b. Insert a key into the keyway, and slide on gear G2.
    c. Fix gear G2 with a clamping collar.
    d. Slide the final bearing onto the input shaft.
    e. Install rotary shaft seal.

    f. Install input shaft assembly into the housing base.
5. Assemble intermediate shaft G3-G4
    a. Onto shaft G3-G4, attach two clamping collars.
    b. Insert keys into keyways on the shaft. Slide on gears G3 and G4 from either end.
    c. Slide on a 0.5-inch spacer and a bearing on each end.
    d. Install shaft G3-G4 into the housing base.
6. For Shafts G5-G6, G7-G8, repeat step 3.
7. For the output shaft, repeat step 2.
8. Check Point: Verify that all the stages spin freely without any binding or resistance.
9. Clean and degrease mating surfaces of the housing base and top using a zero-residue cleaning agent.
10. Apply a thin layer of Loctite 2 Flexible Gasket Sealant to the mating surfaces.
11. Lift the top of the housing into place and slowly lower it into position.
12. Check that all components rotate freely.
13. Using 37 5” long 1”-8 UNC grade 5 hex head bolts, fasten the two halves of the housing together as described in section 9.5.
14. Install oil delivery ducts and fill the reservoir with specified oil (section 9.5)
15. Alight and connect the speed reducer unit with the turbine and generator. Secure the hubs and bolt-on disc couplings.
16. Check that all components rotate freely.
17. Fasten the base in the final position to the foundation as specified in Section 9.5.
18. Apply a coating of white Alkyd-based paint over all exterior surfaces and fasteners on the speed reducer.

11. Maintenance and Inspection Plan

FREQUENCY / INSPECTION	Whichever comes first → / ↓ Every	Revolutions of Generator Shaft † (×10^8 revolutions)			
		1.2	2.4	4.8	9.6
Seals – check for leaks*	6 months		●	●	●
Bolts – check for cracked coating indicating loosening*	6 months	●			
Bolts – retorque	5 years				●
Anchors – retorque	3 years				●
Couplings – check Fastener torque*	6 months	●			
Couplings – check alignment*	5 years			●	●
Coating – check for corrosion or damage	1 year		●	●	●
Coating – replace	12 years				
Lubricant – check level*	1 month	●			
Lubricant – change	1 year	●			
Oil filter – replace	5 years			●	●
Oil sump and reservoir – check for leaks*	1 year	●			
Vibration – check magnitude and log	1 month				
Auditory – check for unexpected grinding, clicking, or whining	6 months	●			
Shafts – check bending and misalignment*	5 years			●	●
End-of-day operating temperature – Check and log	1 month				
Bearings – check *	5 years			●	●

*: Replace, add, adjust, clean, or torque if necessary.

†: For higher revolution readings, repeat inspections at frequency intervals established here.

12. Concluding Remarks

The design laid out in this report describes and justifies the main mechanical components of a mechanical speed reducer suitable for the 15 MW power plant application specified. The system lifetime shall be no more than 25 at rated operating speeds and durations. Any instrumentation, electrical machines, or other ancillary components such as the foundation, oil handling system, and paint coating are reviewed and handled by professionals in those respective fields. The next steps recommended for the manufacture and implementation of this design is to contact an iron foundry and a heavy machine shop to determine a cost and timeline for manufacturing the housings, gears, and shafts.

Appendix A – Extra Calculations

Figure A.1 Ratio Calculations:

1. Stages 1 and 2:

$$e_{3/2} = -\frac{N2}{N3} = -2.7 \rightarrow N2 = 2.7N3$$

$$e_{5/4} = -\frac{N4}{N5} = -2.7 \rightarrow N4 = 2.7N5$$

2. Stages 3 and 4:

$$e_{\frac{7}{6}} = -\frac{N6}{N7} = -1.35 \rightarrow N6 = 1.35N7$$

$$e_{9/8} = -\frac{N8}{N9} = -1.35 \rightarrow N8 = 1.35N9$$

Figure A.2 Se Calculation For SN Diagram For Gears:

Correction Factor	Formula	Value
C_{load}	Bending	1
C_{size}	d > 10in	0.6
$C_{surface}$	A*(250[kpsi])^b	2.22
C_{temp}	T < 840°F	1
C_{reliab}		0.85
Se'	100 [kpsi]	100

Used the table values above to calculate the corrected Se value:

$$Se = Cload * Csize * Csurface * Ctemp * Creliab * Se' = 113.22\ [kpsi]$$

Appendix B – Shafts, Spacers, Key, Locking Collar

Figure B.1 Input Shaft

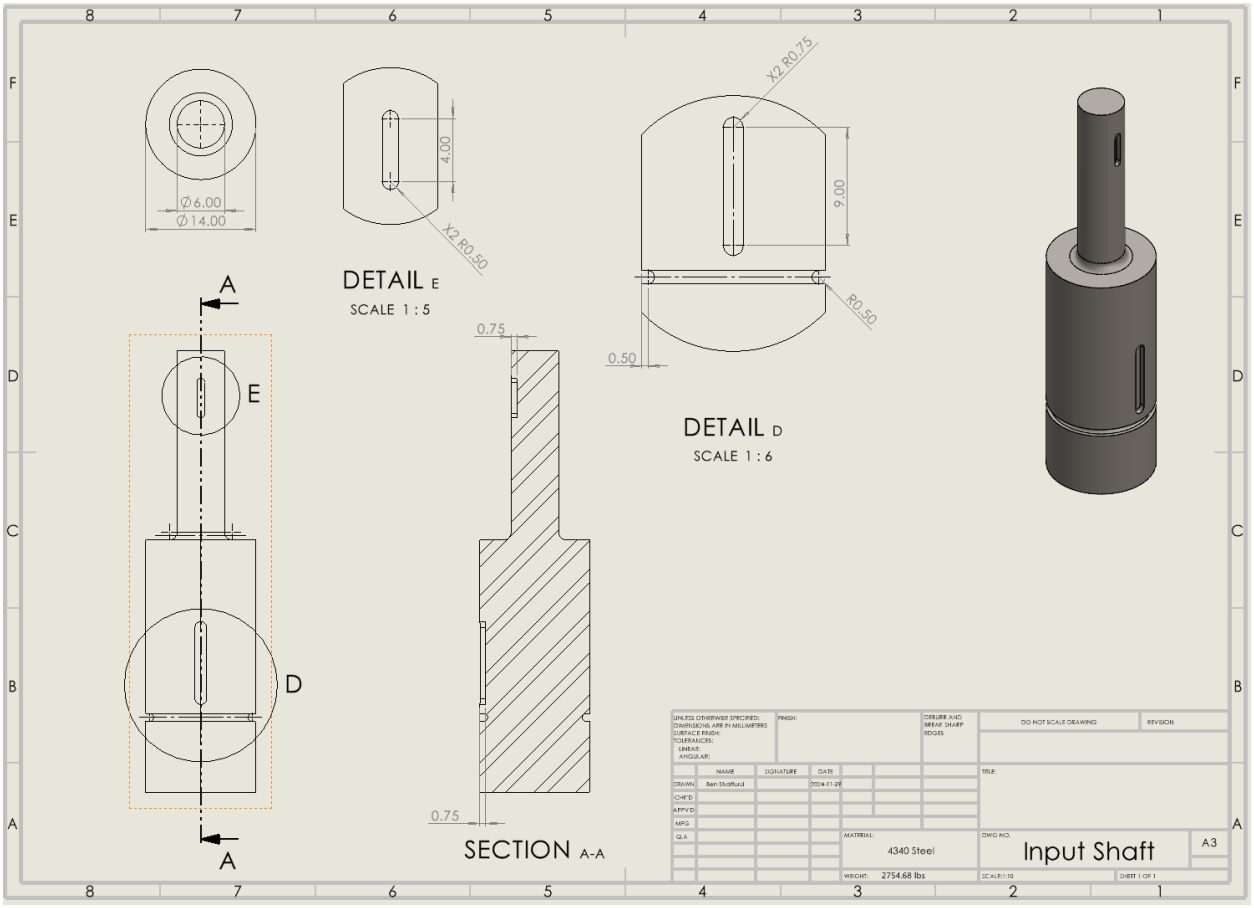


Figure B.2 Intermediate Shaft

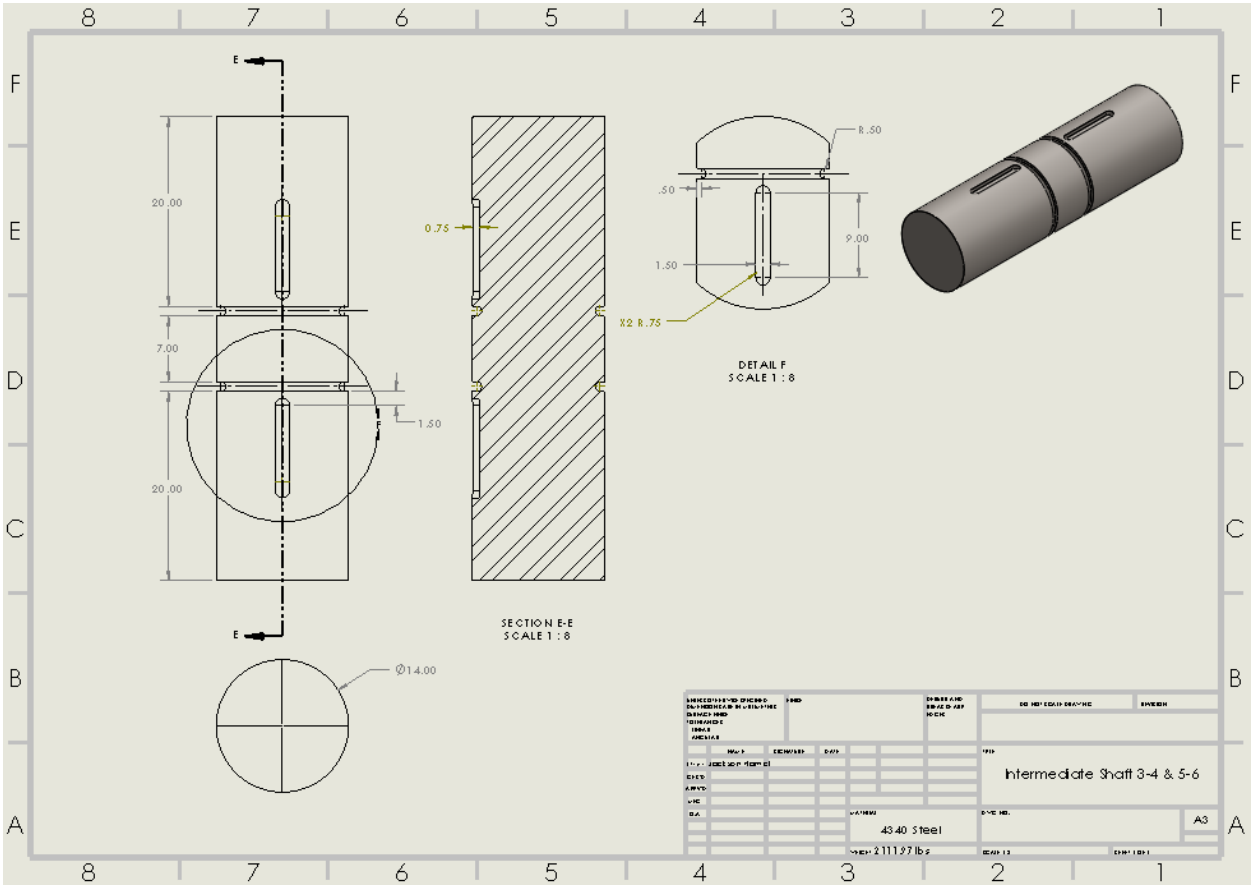


Figure B.3 Intermediate Shaft 2

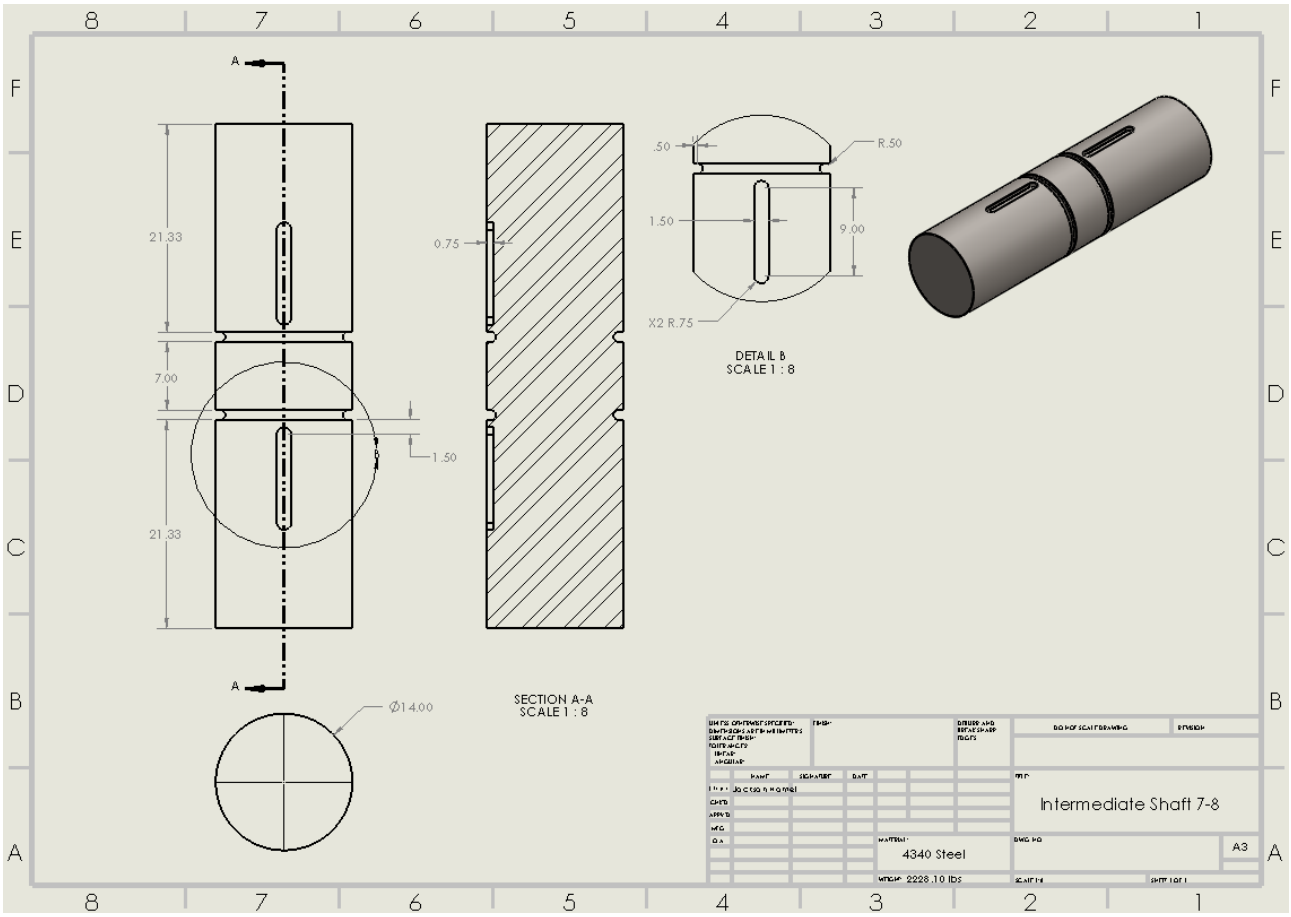


Figure B.4 Output Shaft

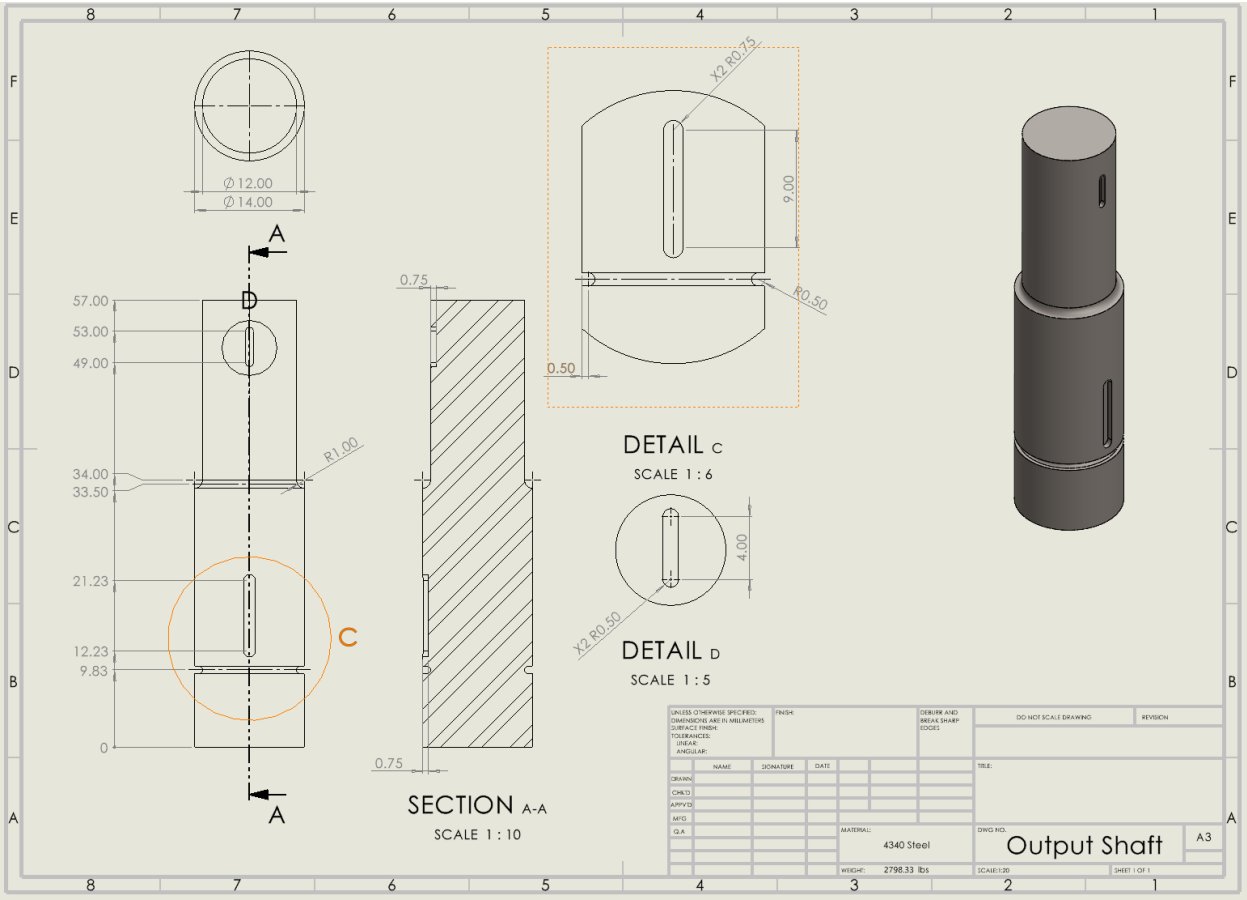


Figure B.5 2” Spacer

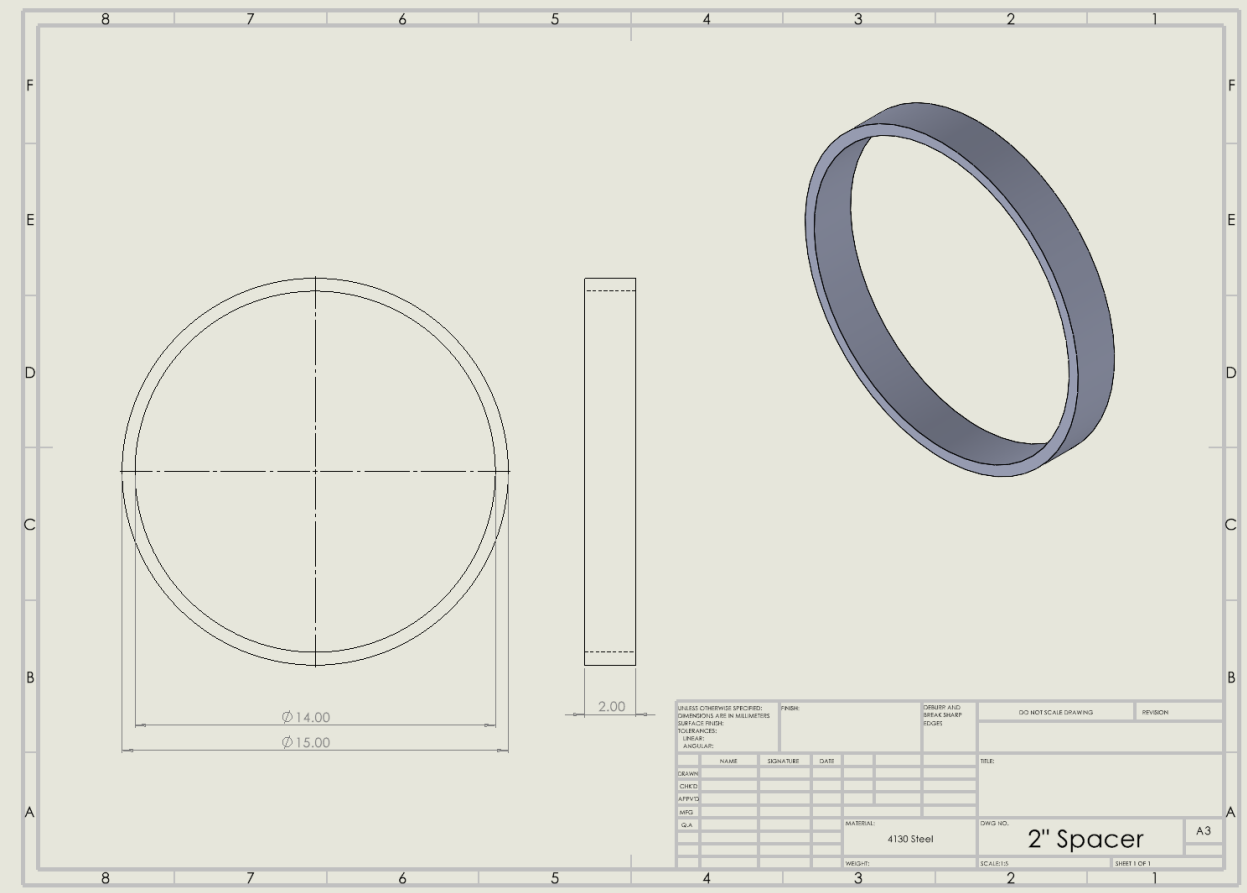


Figure B.6 ½" Spacer

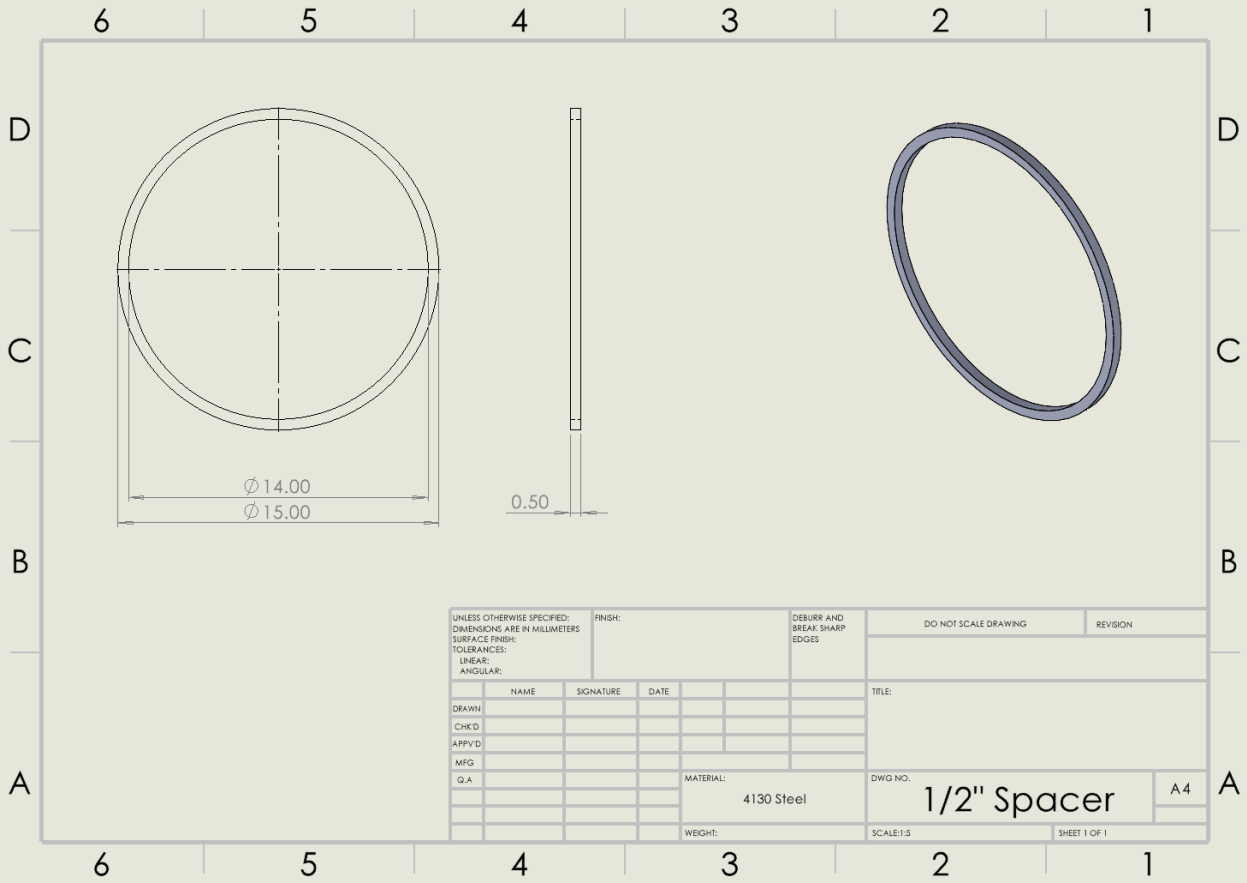


Figure B.7 Key

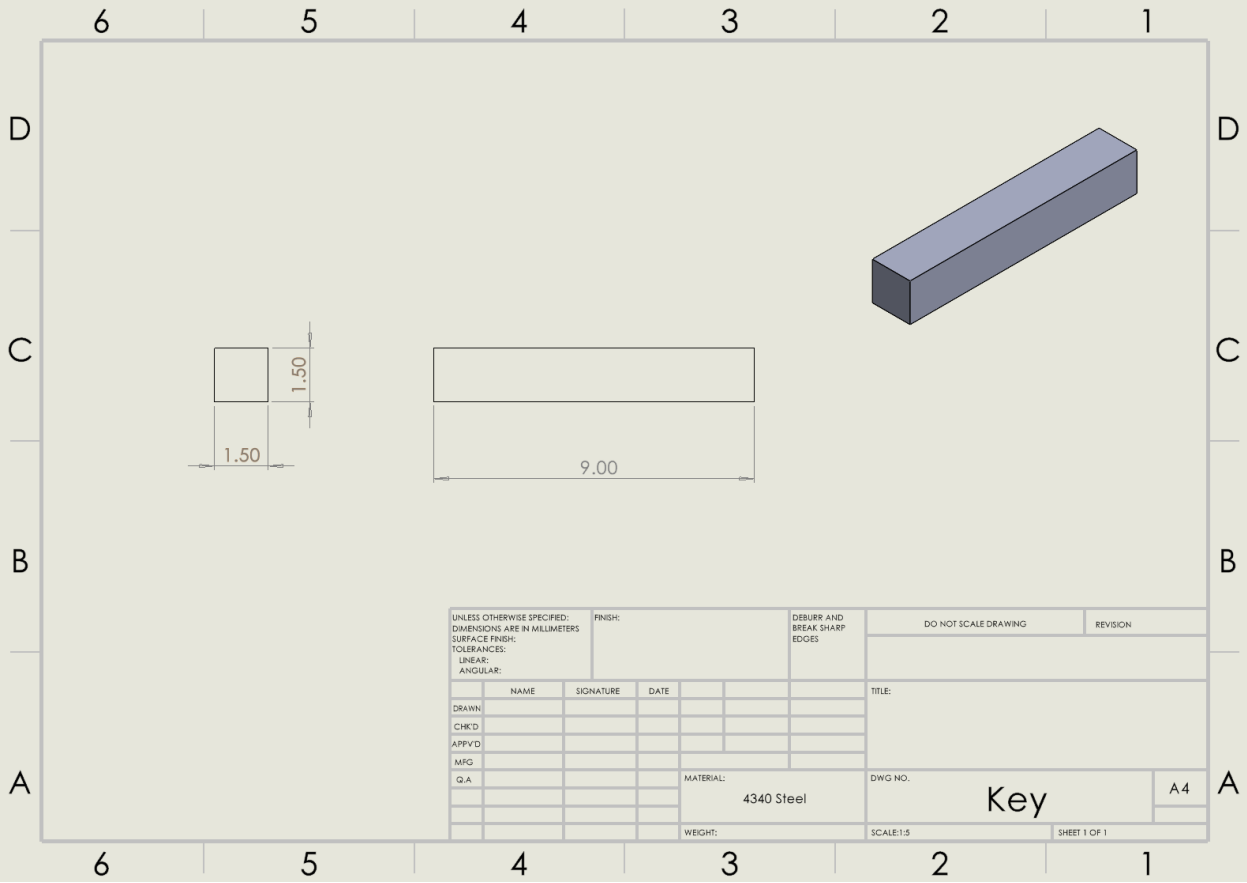
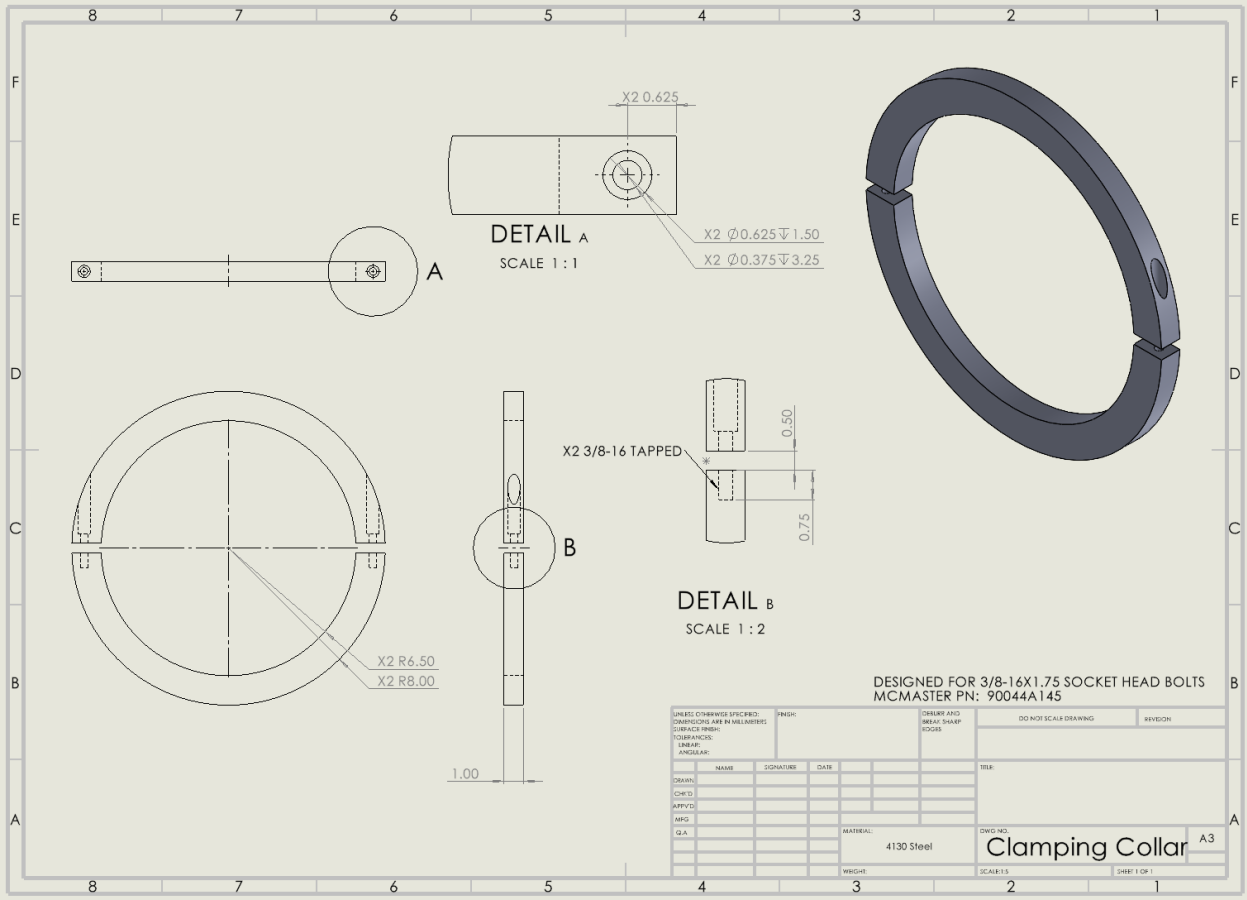


Figure B.8 Clamping Collar



Appendix C - Gears

Figure C.1 Gears 2 and 4

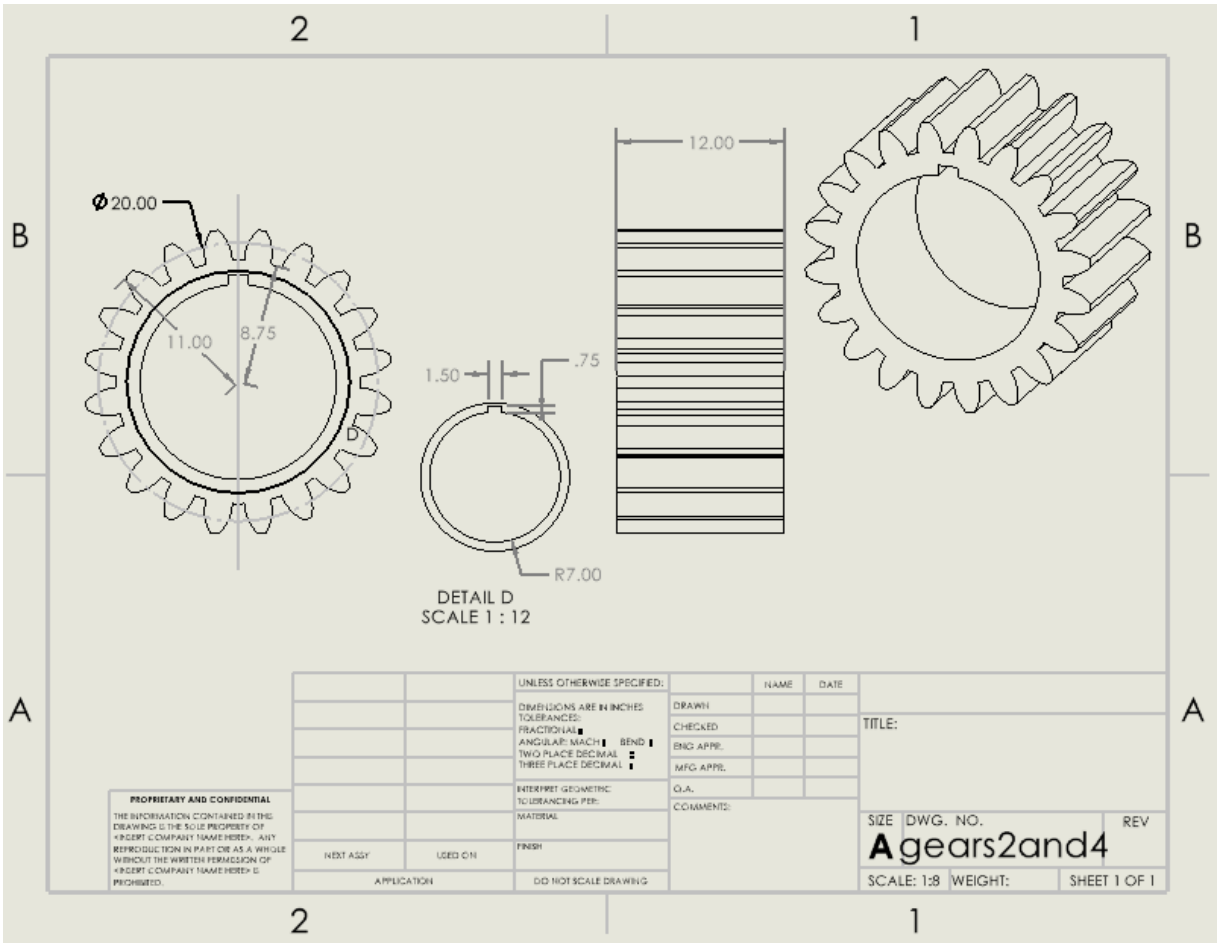


Figure C.2 Gears 3 and 5

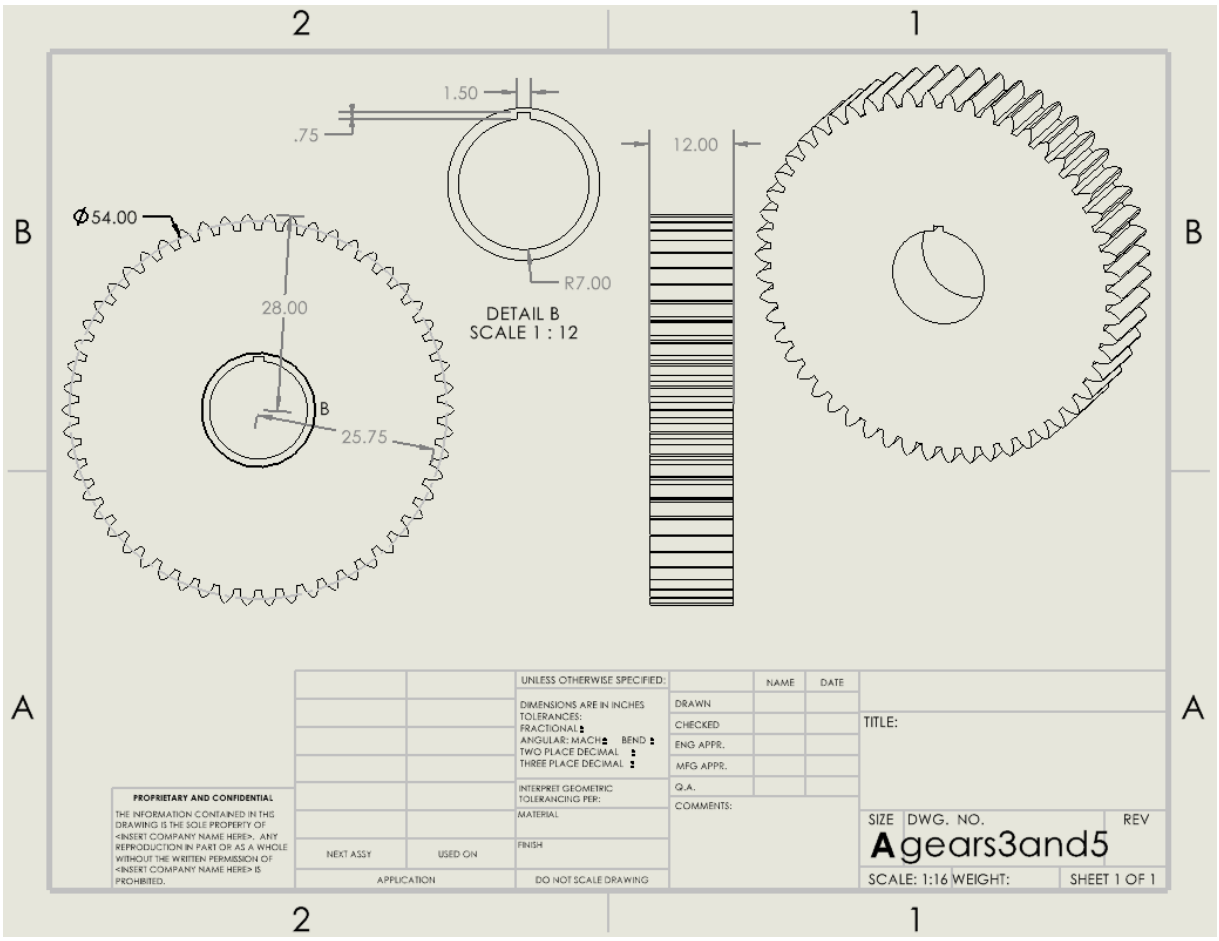


Figure C.3 Gears 6 and 8

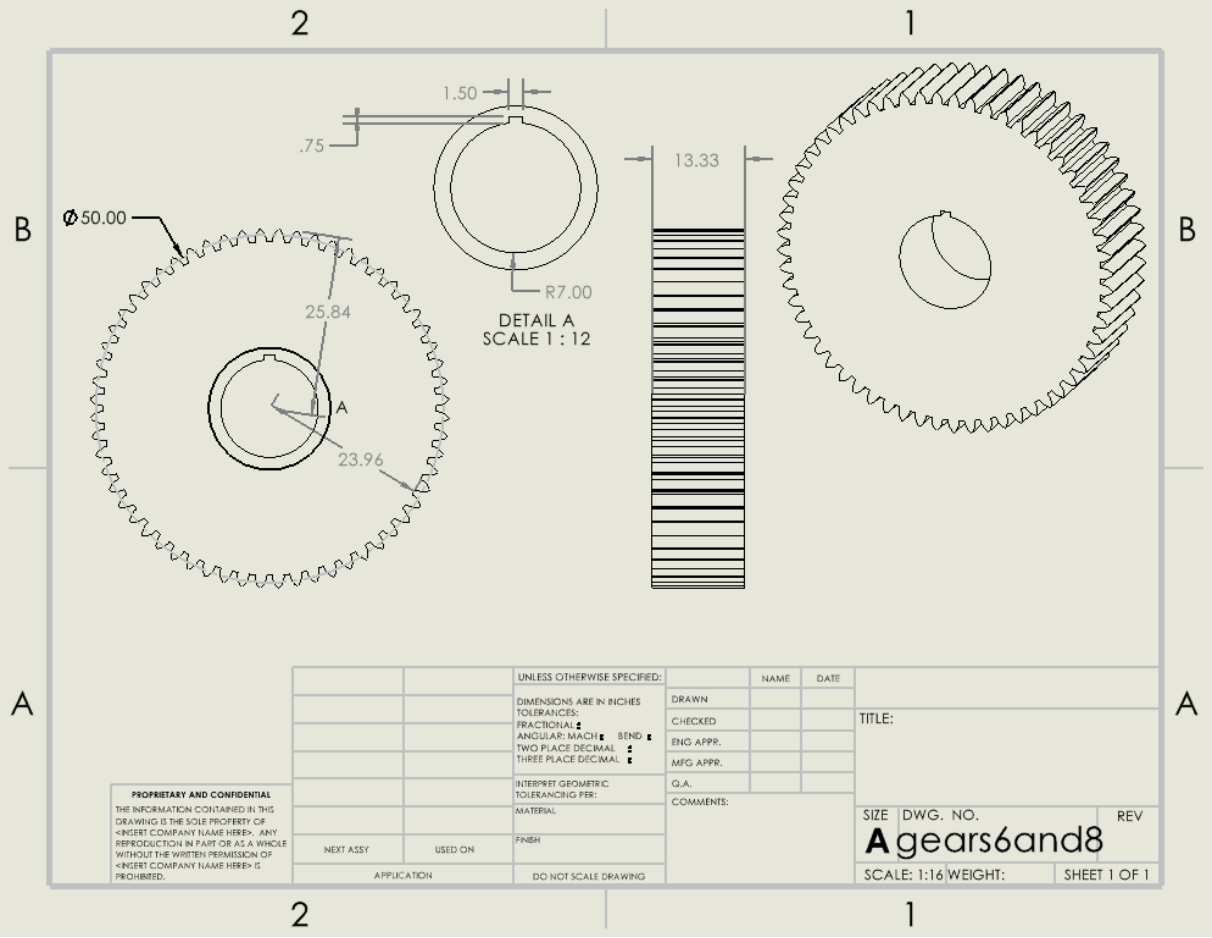
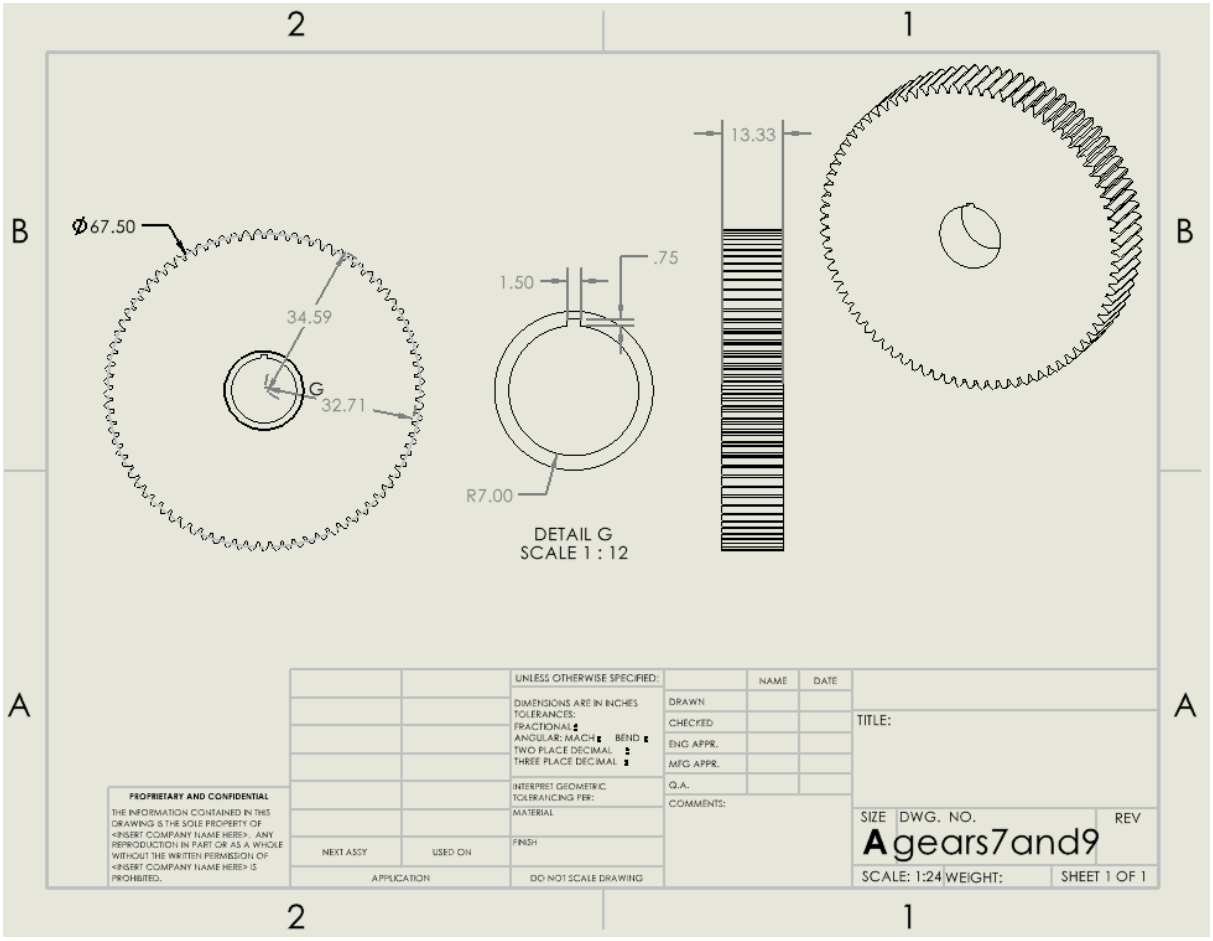


Figure C.4 Gears 7 and 9



Appendix D - Assembly, Housing, and Fasteners

Figure D.1 Gearbox Assembly

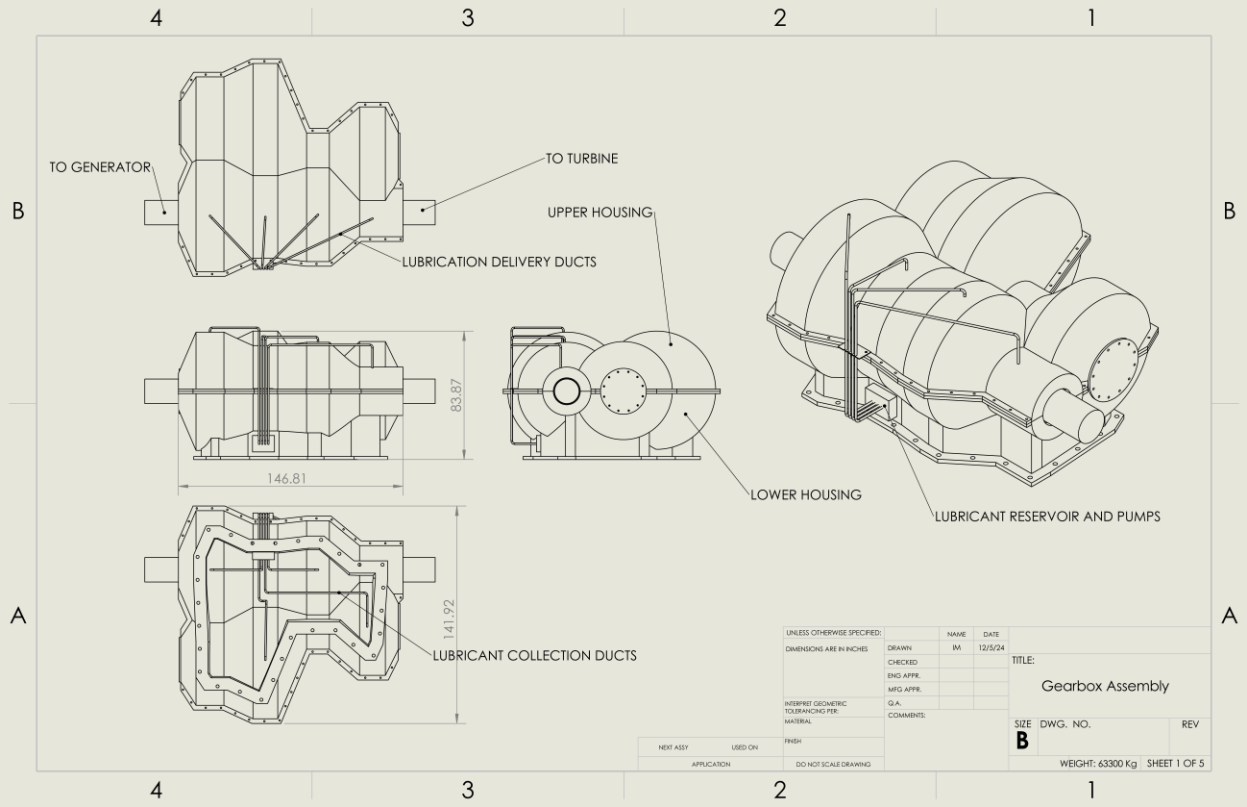


Figure D.2 Gearbox Assembly Exploded View

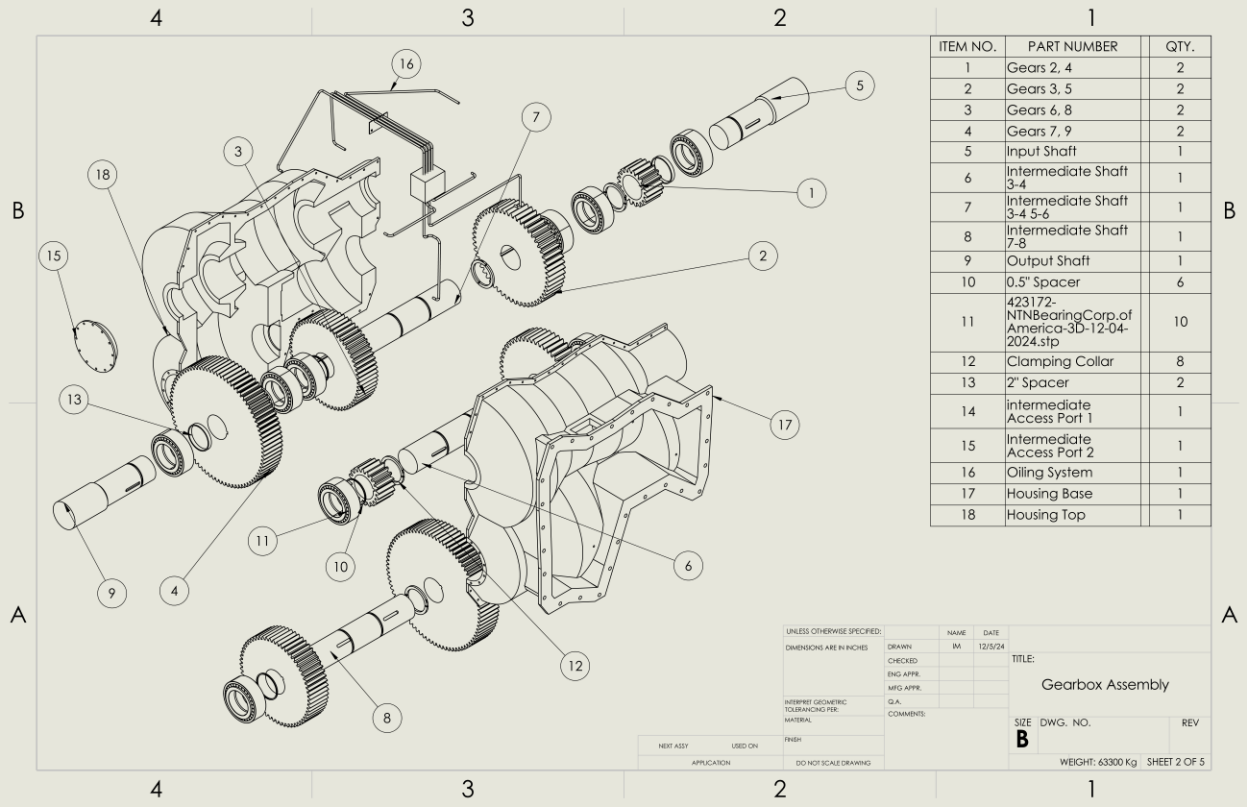


Figure D.3 Gearbox Section View 1

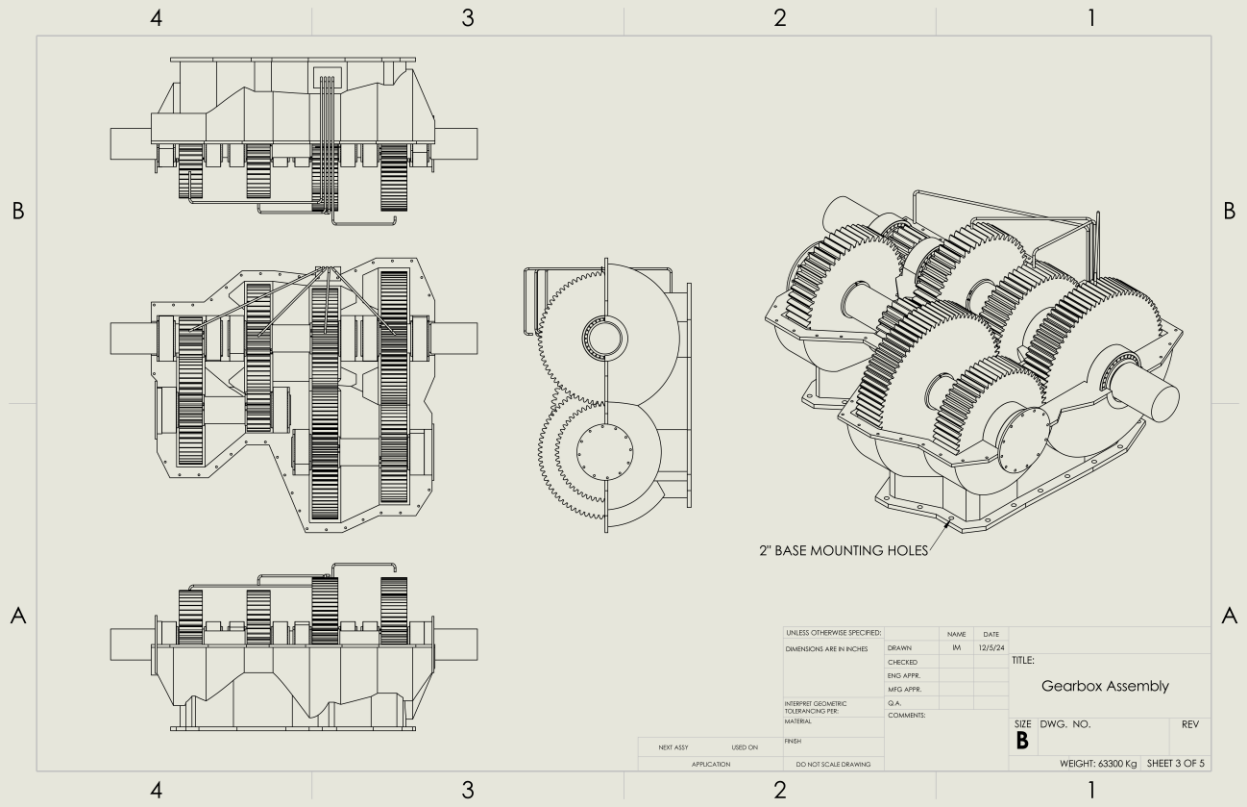


Figure D.4 Gearbox Section View 1

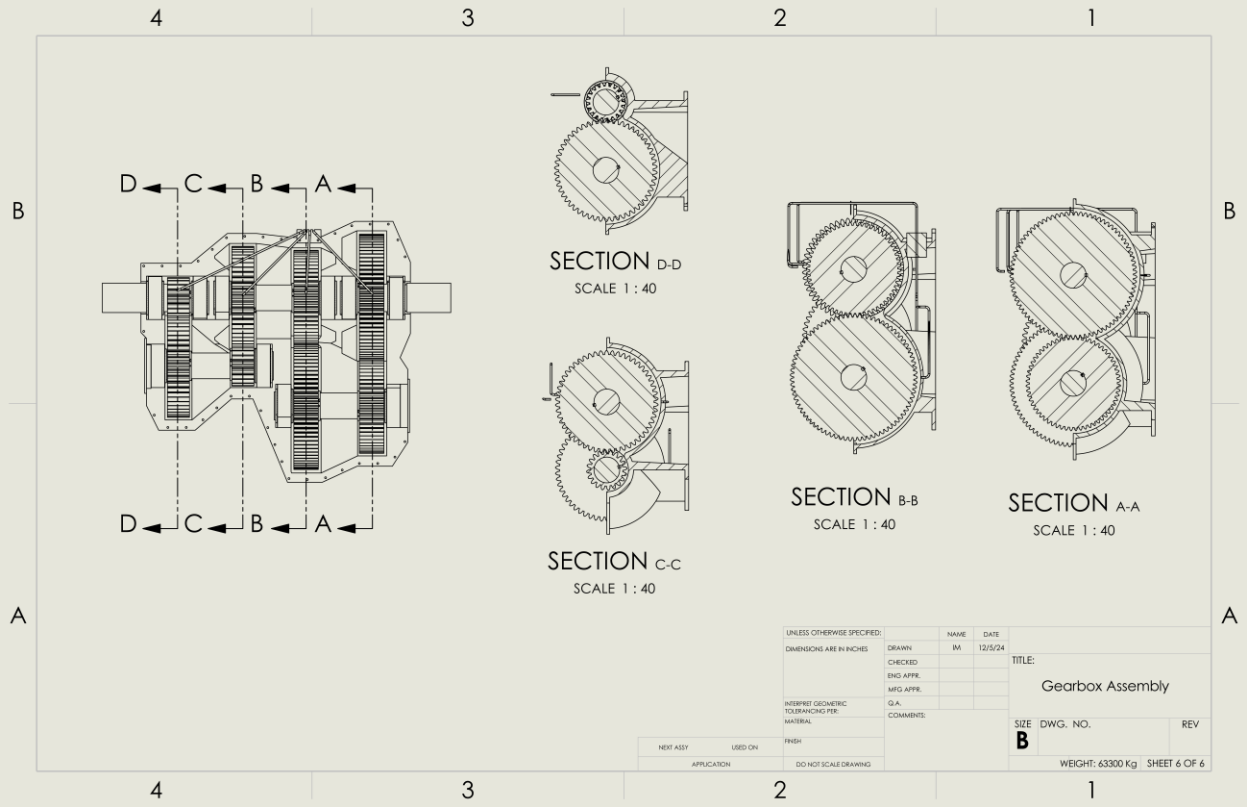


Figure D.5 Housing Base

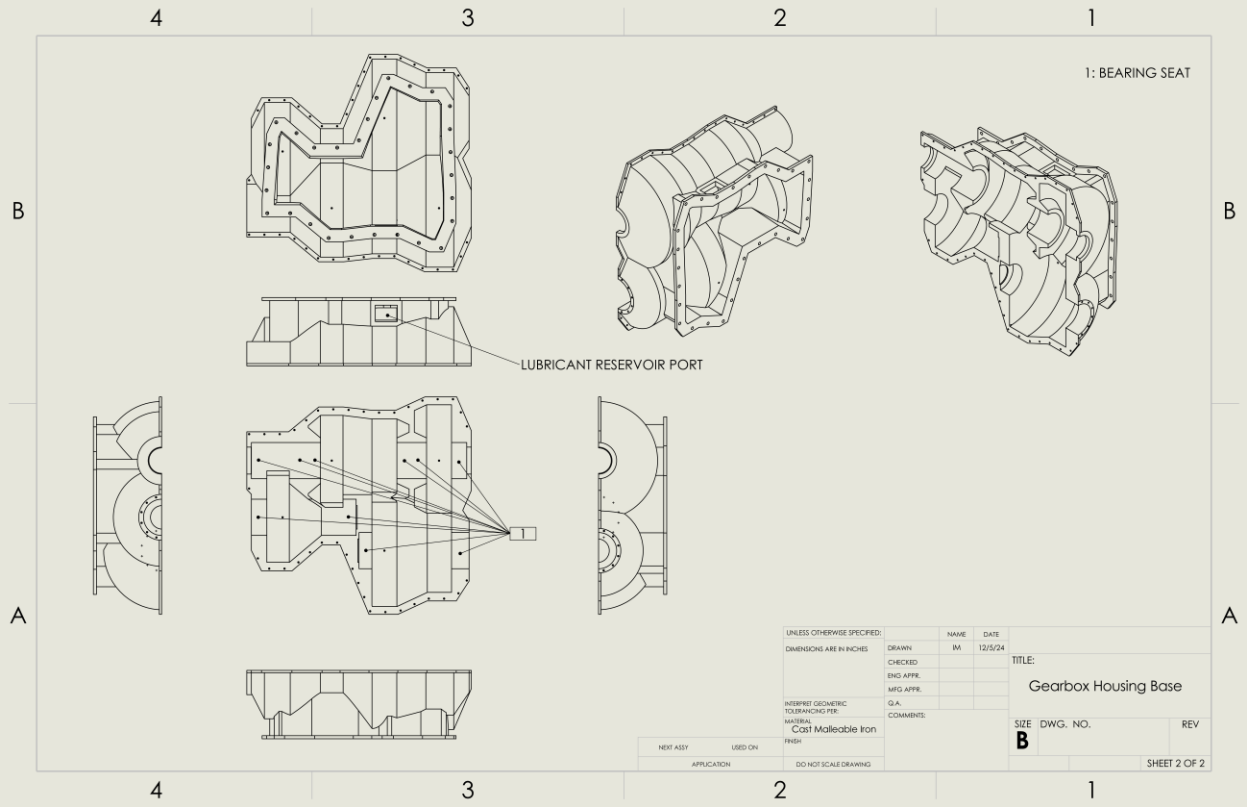


Figure D.6 Housing Base Section Views

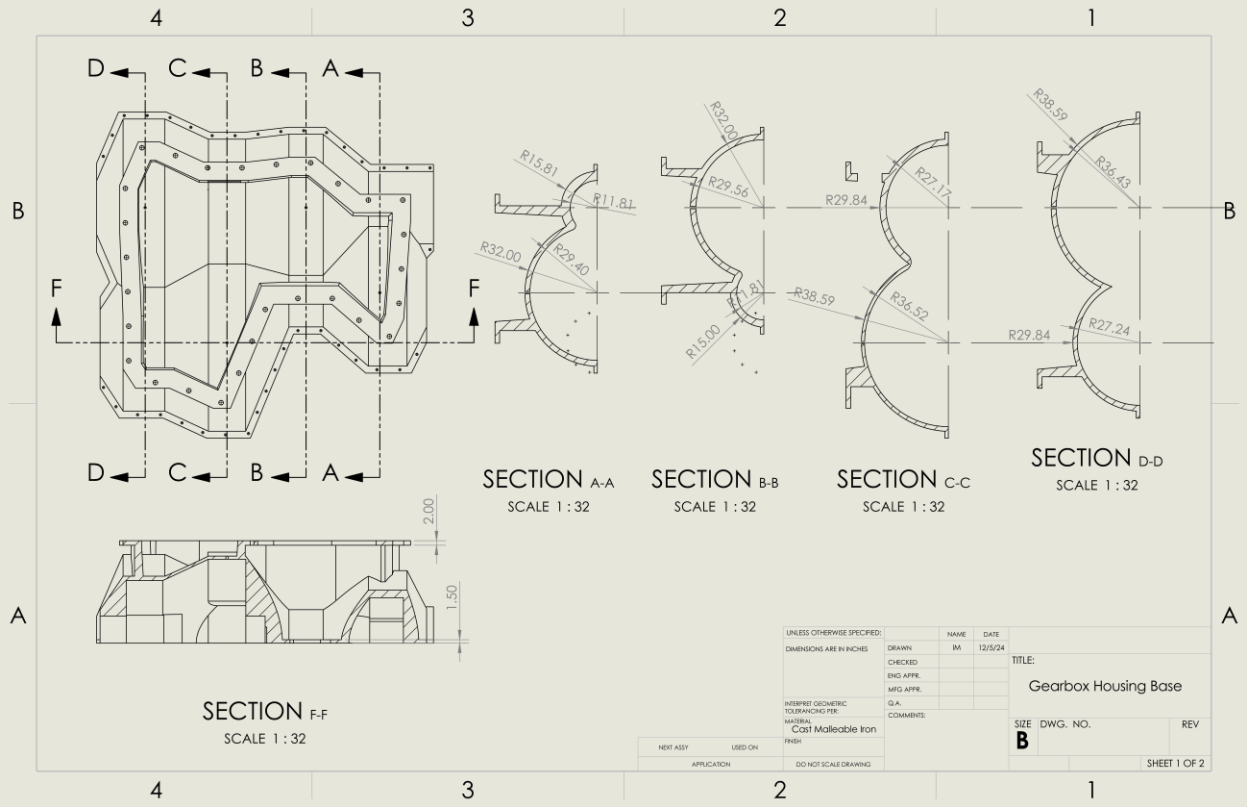


Figure D.7 Housing Top

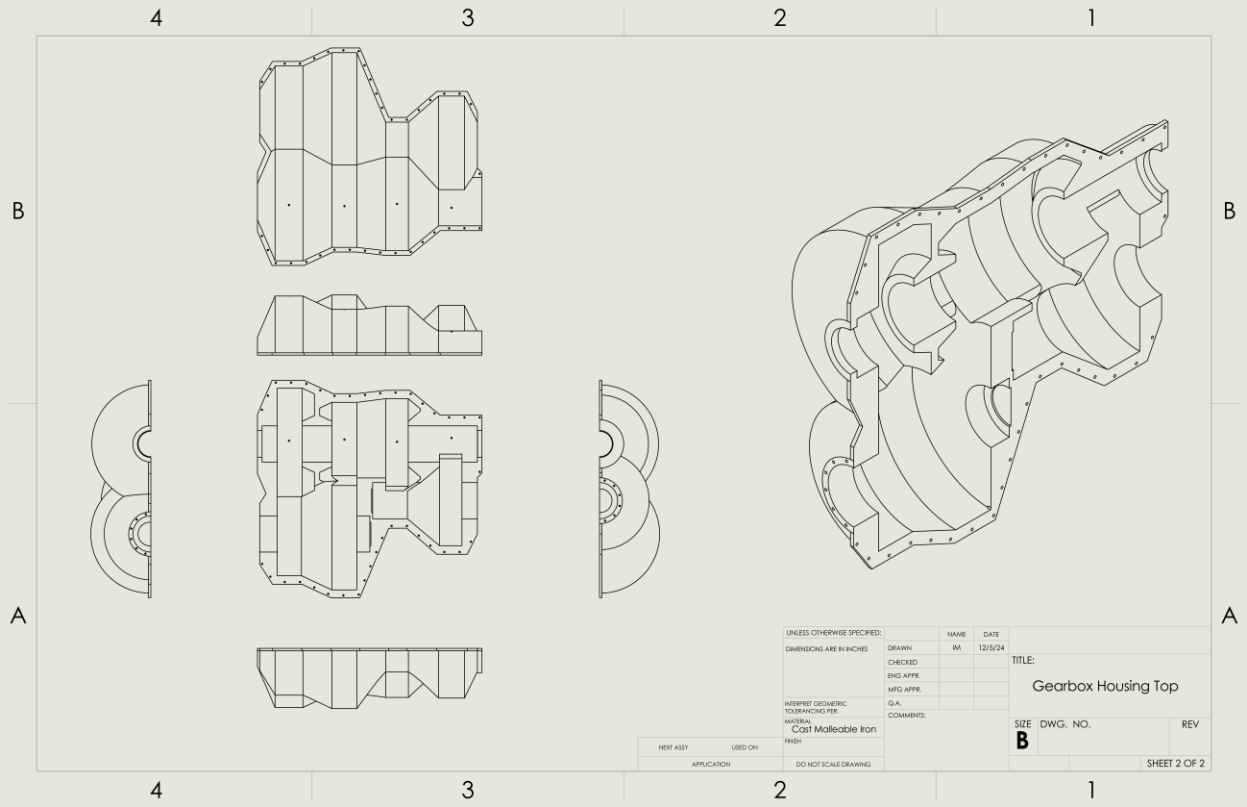


Figure D.8 Housing Top Section Views

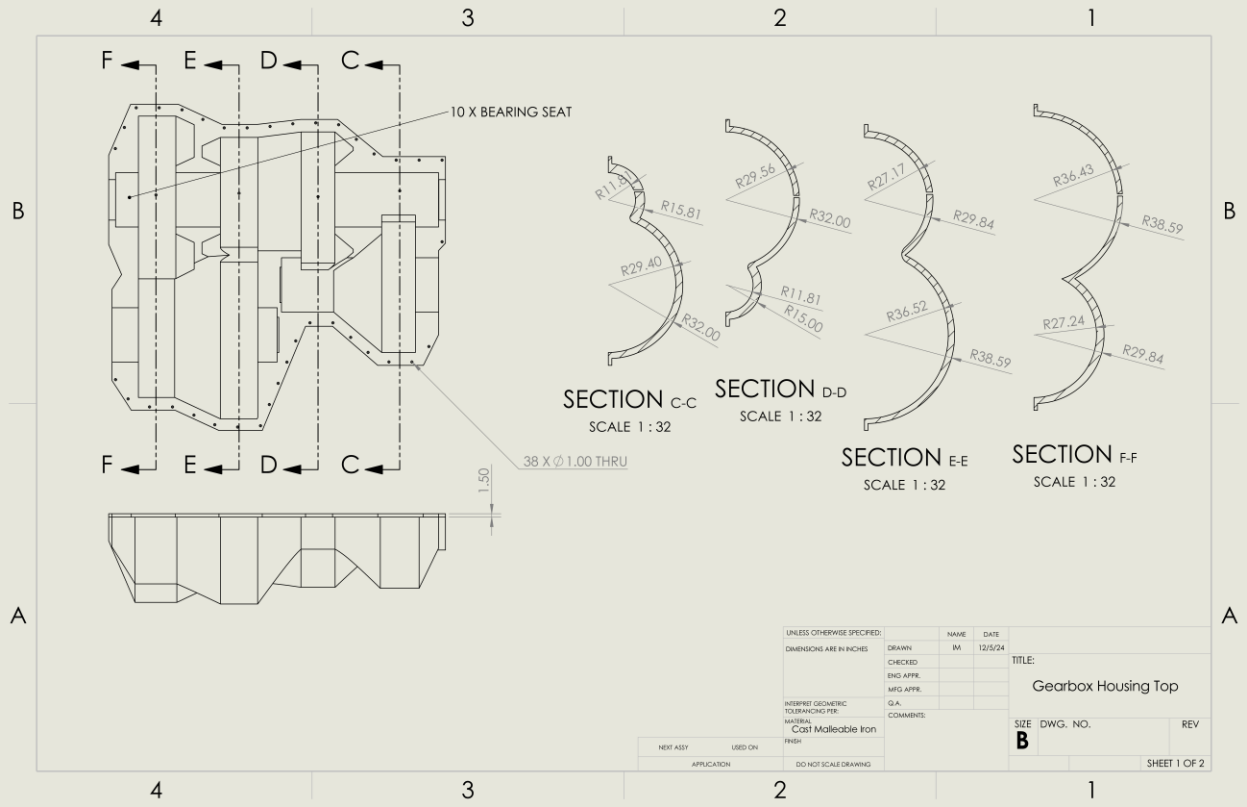


Figure D.9 McMaster Fastener 97110A701 [17]

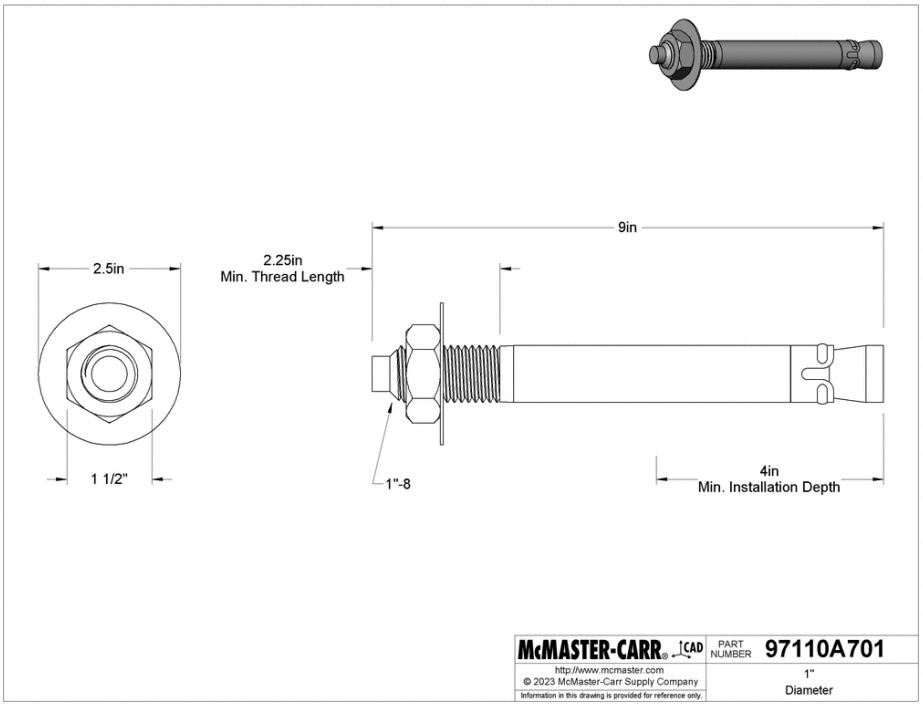


Figure D.2.1 McMaster Fastener 91268A966 [17]

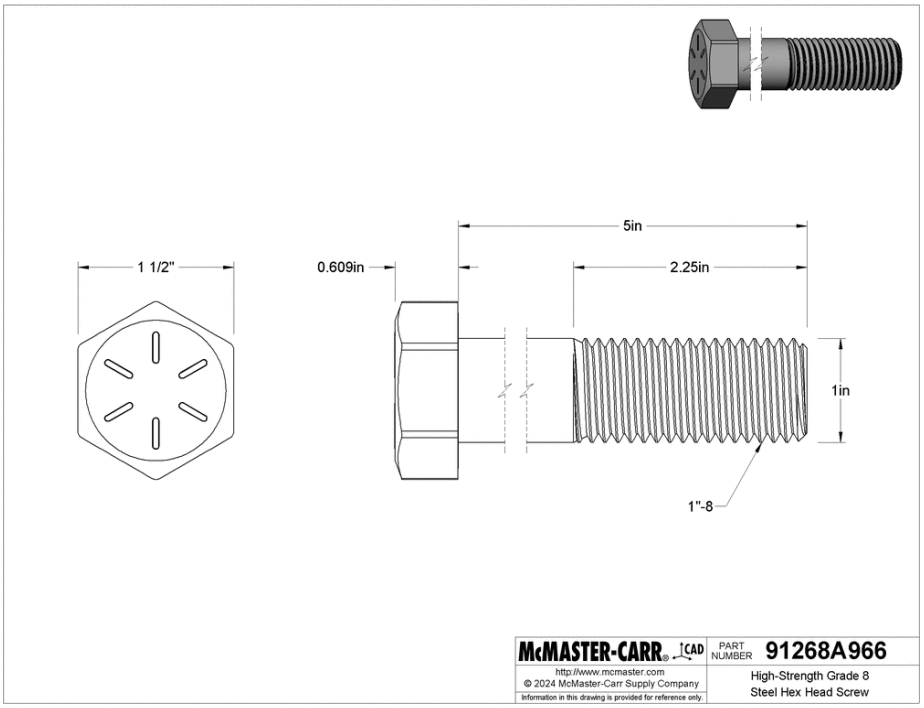
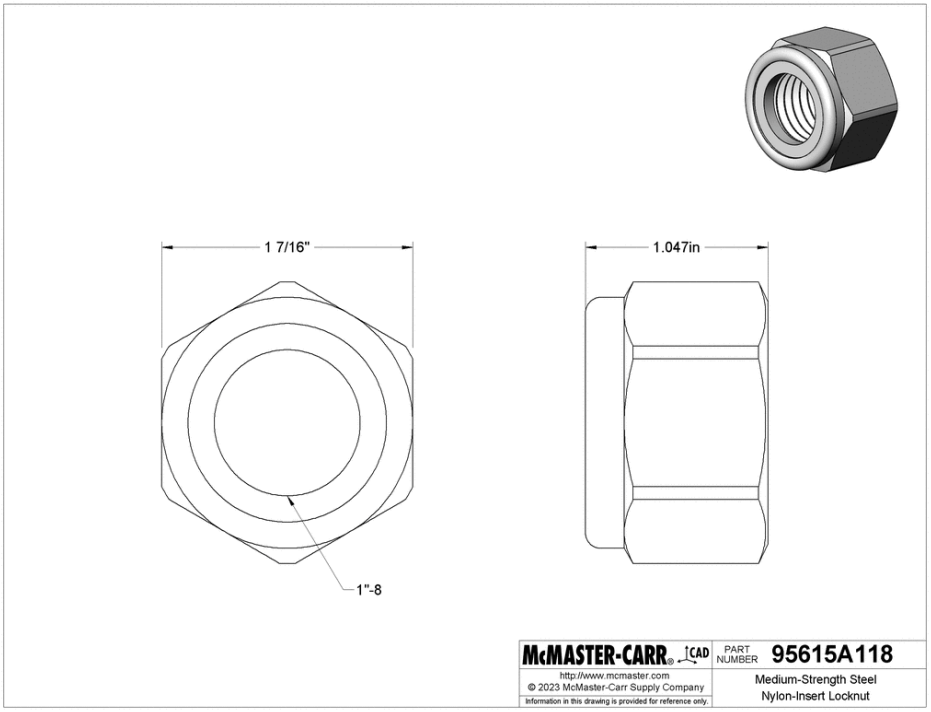


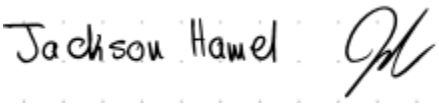
Figure D.2.2 McMaster Locknut 95615A118 [17]



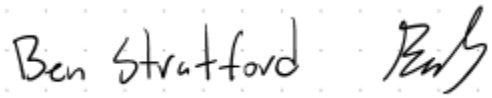
Appendix E – Teams Expectations Agreement

1) Each member will be individually accountable for their work.
2) Each member will do their best to attend group meetings.
3) Team members should communicate regularly, keeping other members updated.

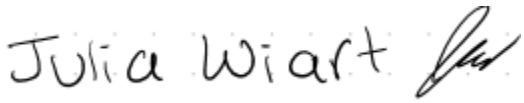
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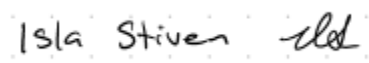
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Name:



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